

Building Holographic Entanglement by Measurement

Brian Swingle (Brandeis)

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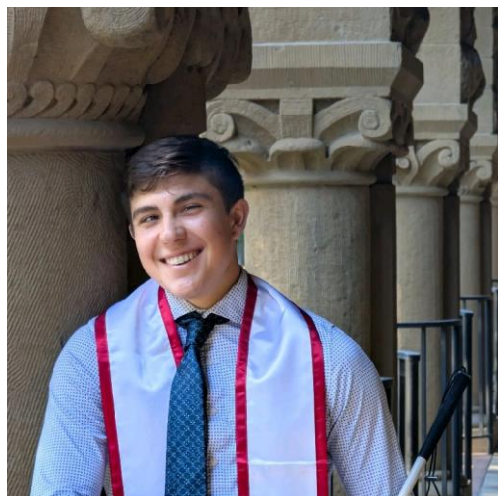
Midjourney-S



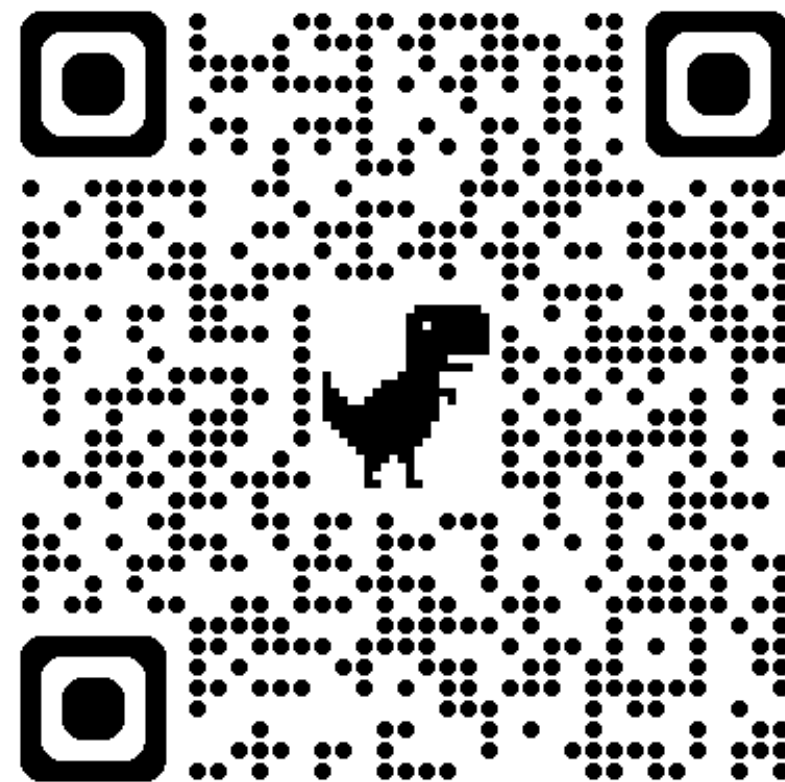
Jonathan Jeffrey



Monika Schleier-Smith

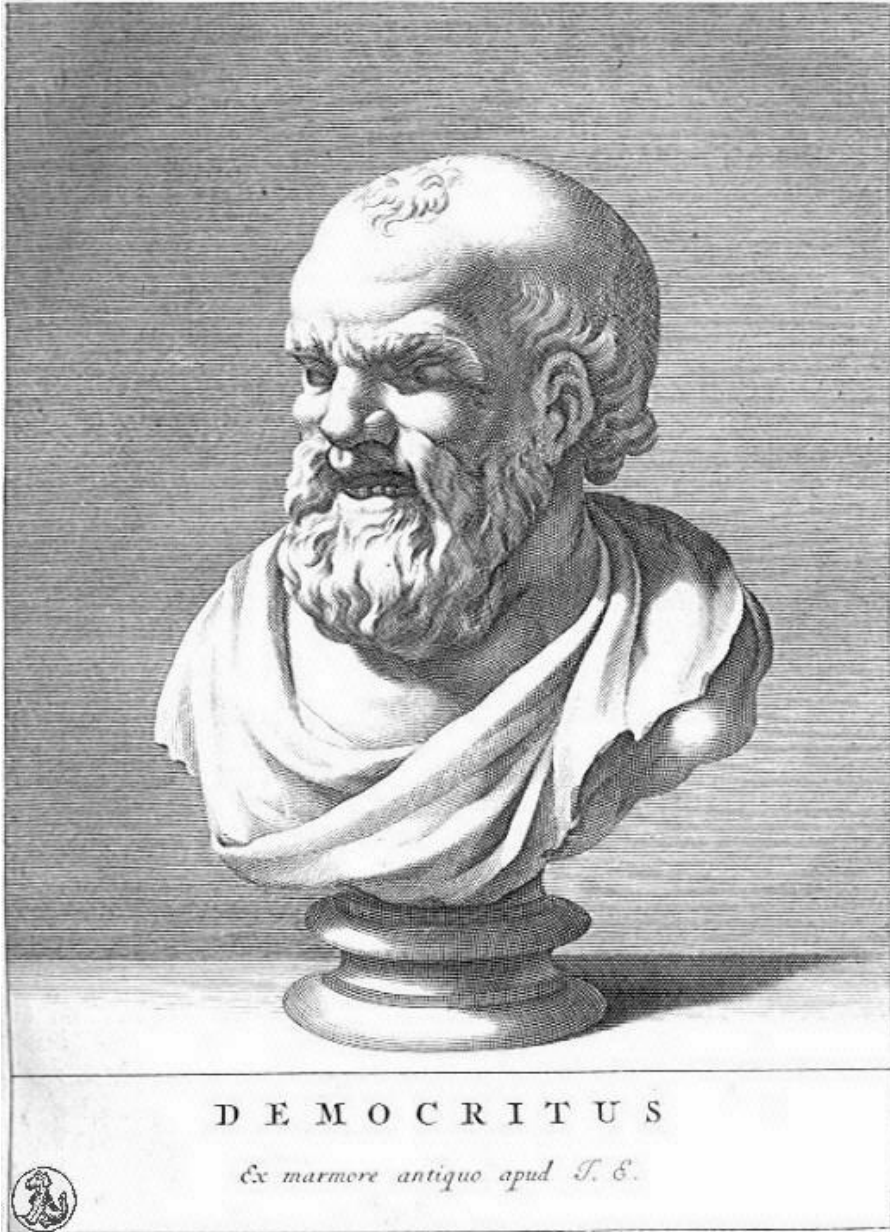


Lucien Gandarias



physicsmonkey.github.io

The problem of quantum gravity



“By convention sweet and by convention bitter, by convention hot, by convention cold, by convention color; but in reality atoms and void.”

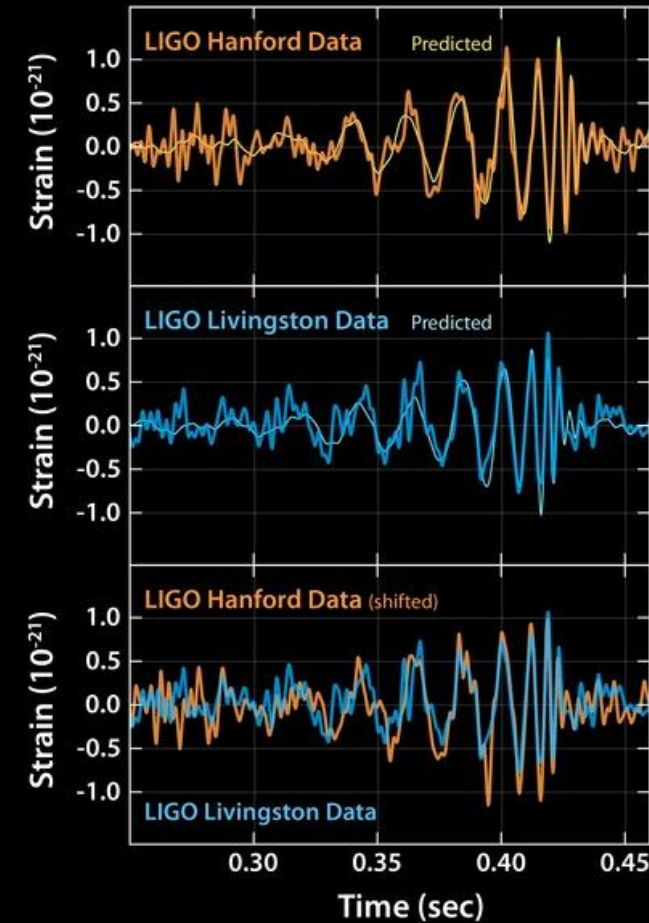
Why is the void interesting?

Quantum gravity is mysterious!

$$S = \frac{\text{Area}}{4G_N\hbar} \quad [\text{Bekenstein, Hawking}]$$



[Supermassive black hole in M87 – EHT, 2019]



$$\tau \sim \frac{\hbar}{T}$$

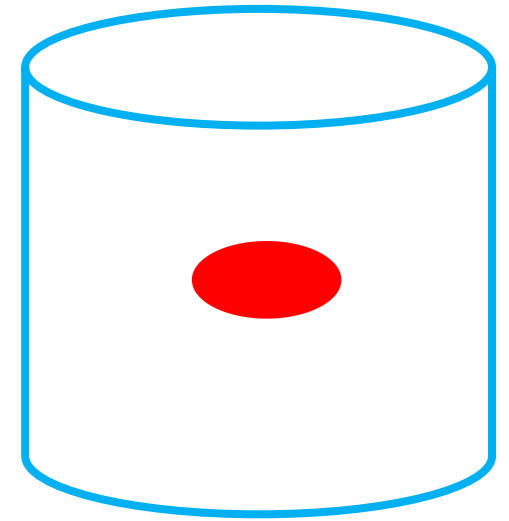
What are the “atoms” of spacetime and how do they yield the area/entropy relationship?

A toy model

AdS/CFT and “it from qubit”

For a “universe in a bottle” (Anti de Sitter space), the AdS/CFT correspondence gives a framework for quantum gravity: [Maldacena, ...]

- Wonderful theoretical lab to study black holes, strong coupling dynamics, hydrodynamics, ...
- Gravitational physics is intimately linked with quantum information: entanglement and geometry, complexity and black holes, ...
- By simulating the boundary system in a lab, we can hope to experimentally study holography!



Stuff in AdS $\leftrightarrow$
States in dual CFT

Big picture goal

Study models of holographic quantum gravity using table-top atomic, molecular, and optical (AMO) style experiments

- Asking how to do this, even just in principle, is illuminating
- Working within a specific experimentally motivated set of constraints is stimulating
- Ultimately, we want to carry out the experiments and learn!

Examples: lattice QFT, classical chaos, scrambling, ...

Clues: holographic nature of gravity

1. The Hamiltonian is measurable at infinity

$$F = \frac{G m_{\text{test}}}{r^2} \left(\frac{E}{c^2} \right) \quad [\text{Newton}]$$

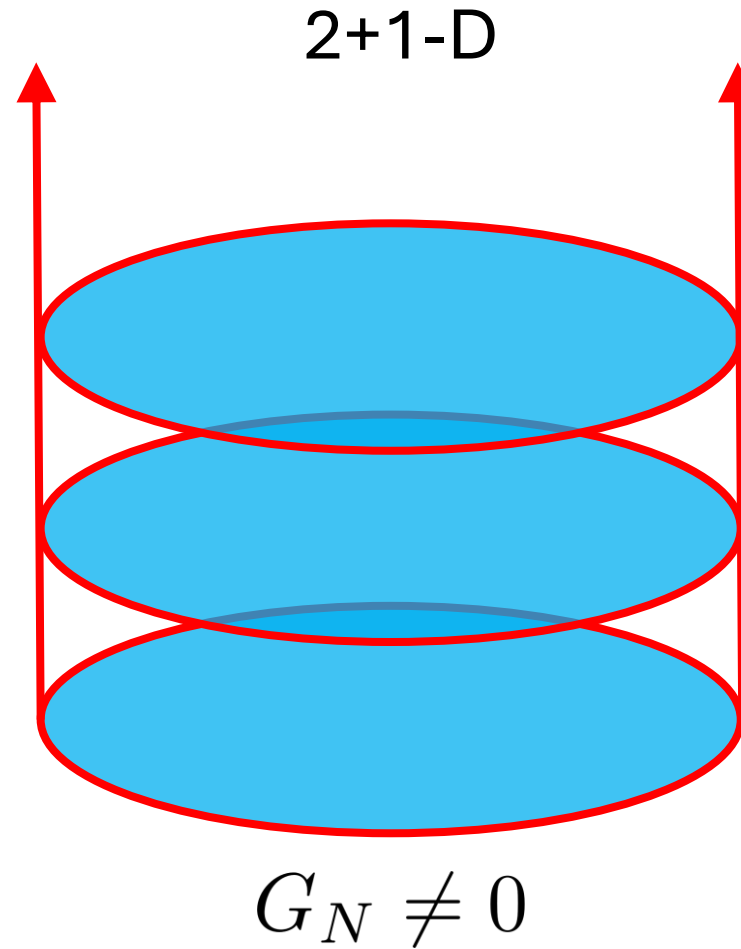
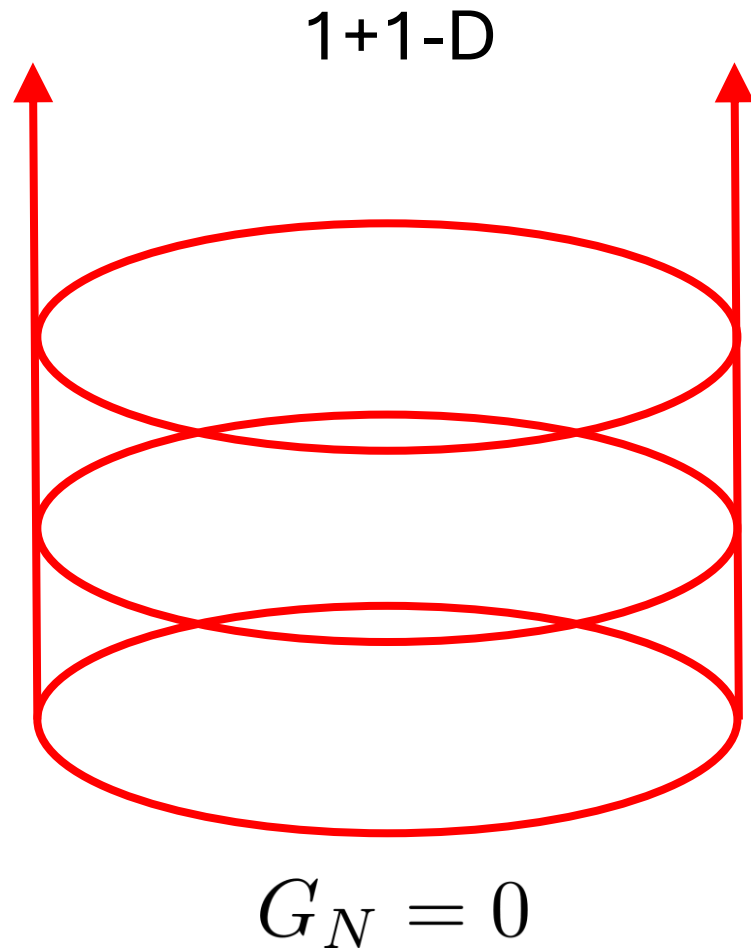
2. Black holes have entropy proportional to their area

$$S_{BH} = \frac{c^3 A_{BH}}{4\hbar G_N} \quad [\text{Bekenstein-Hawking}]$$

3. Gravitational redshift

$$E_{\infty} = E_0 \sqrt{1 - \frac{2GM}{rc^2}} \quad [\text{Einstein}]$$

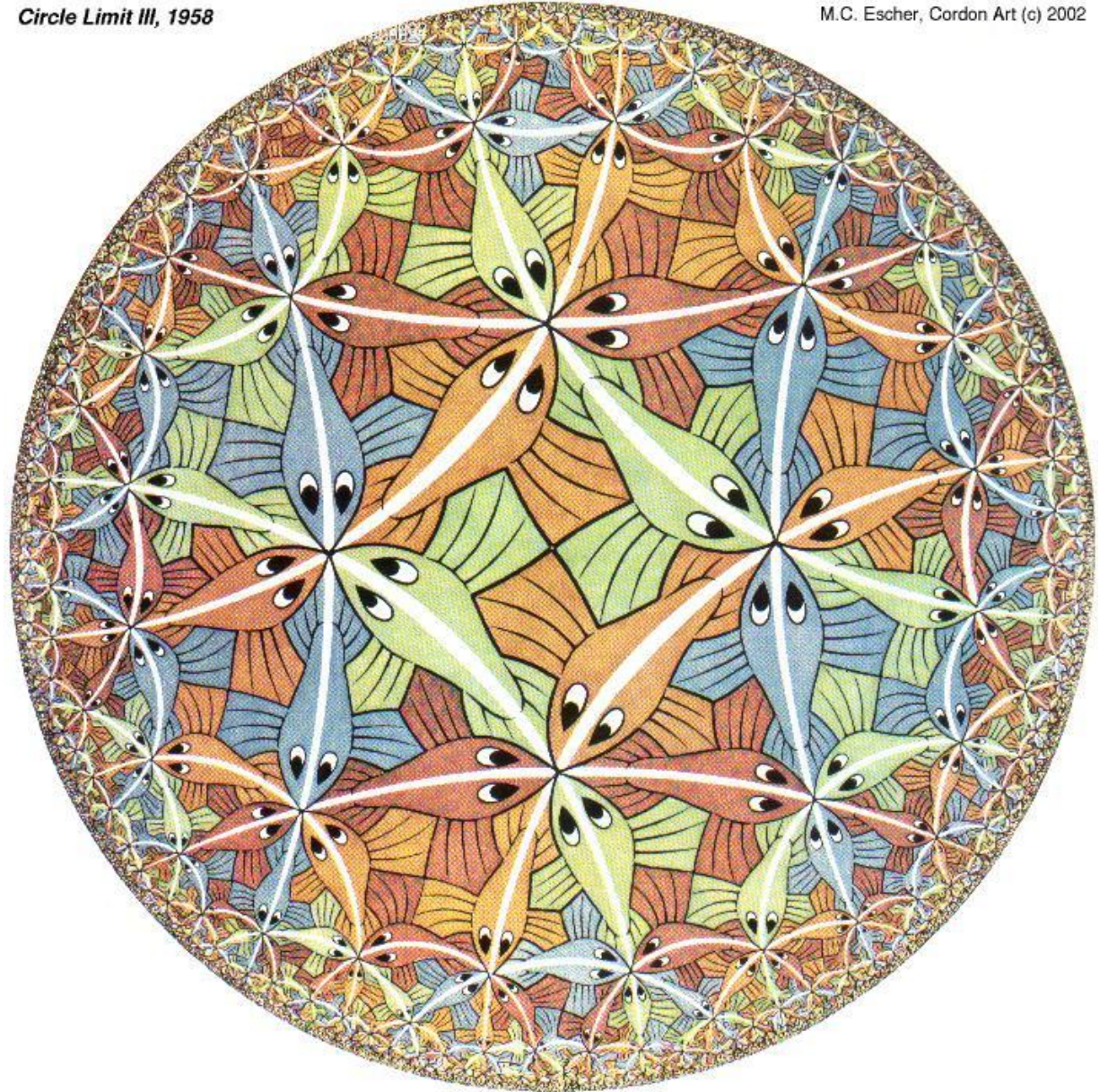
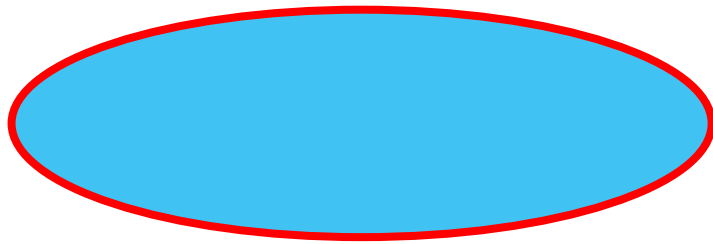
Anti de Sitter space / conformal field theory (AdS/CFT or holographic duality)



[Maldacena]

Circle Limit III, 1958

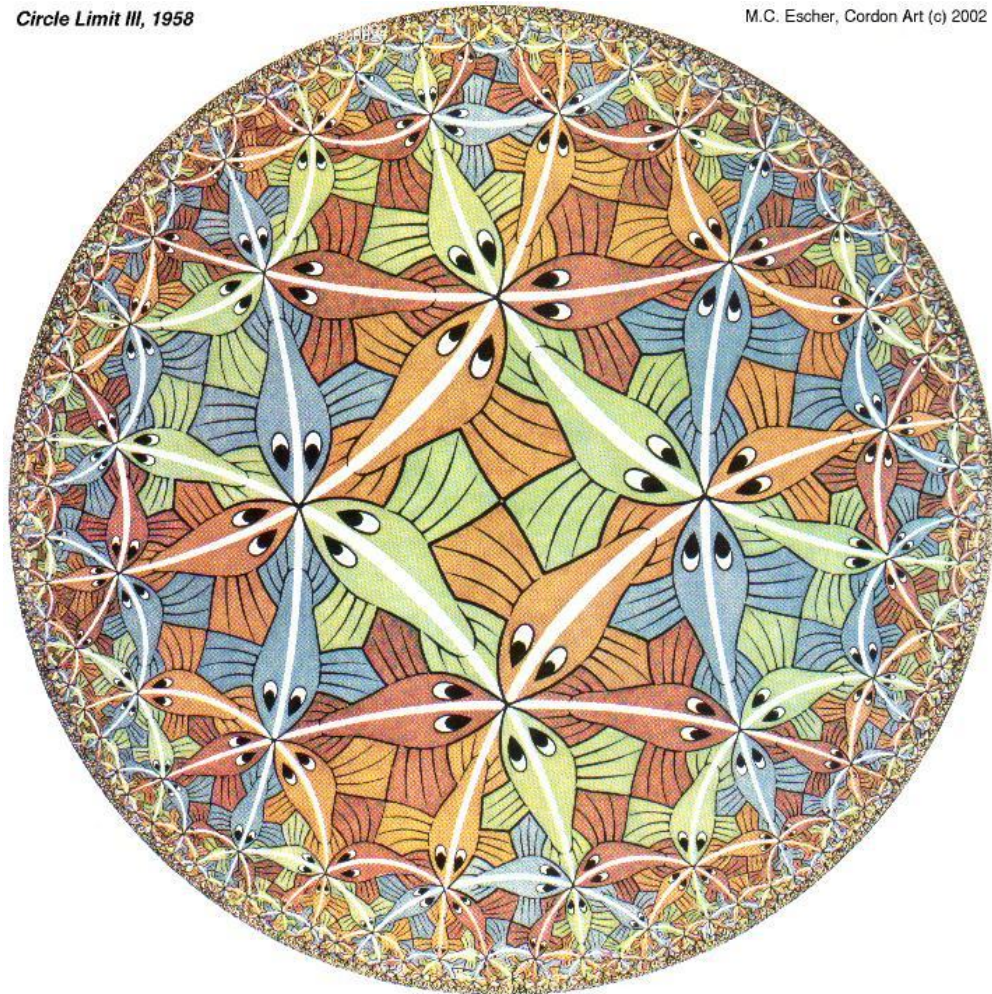
M.C. Escher, Cordon Art (c) 2002



AdS and an AdS black hole

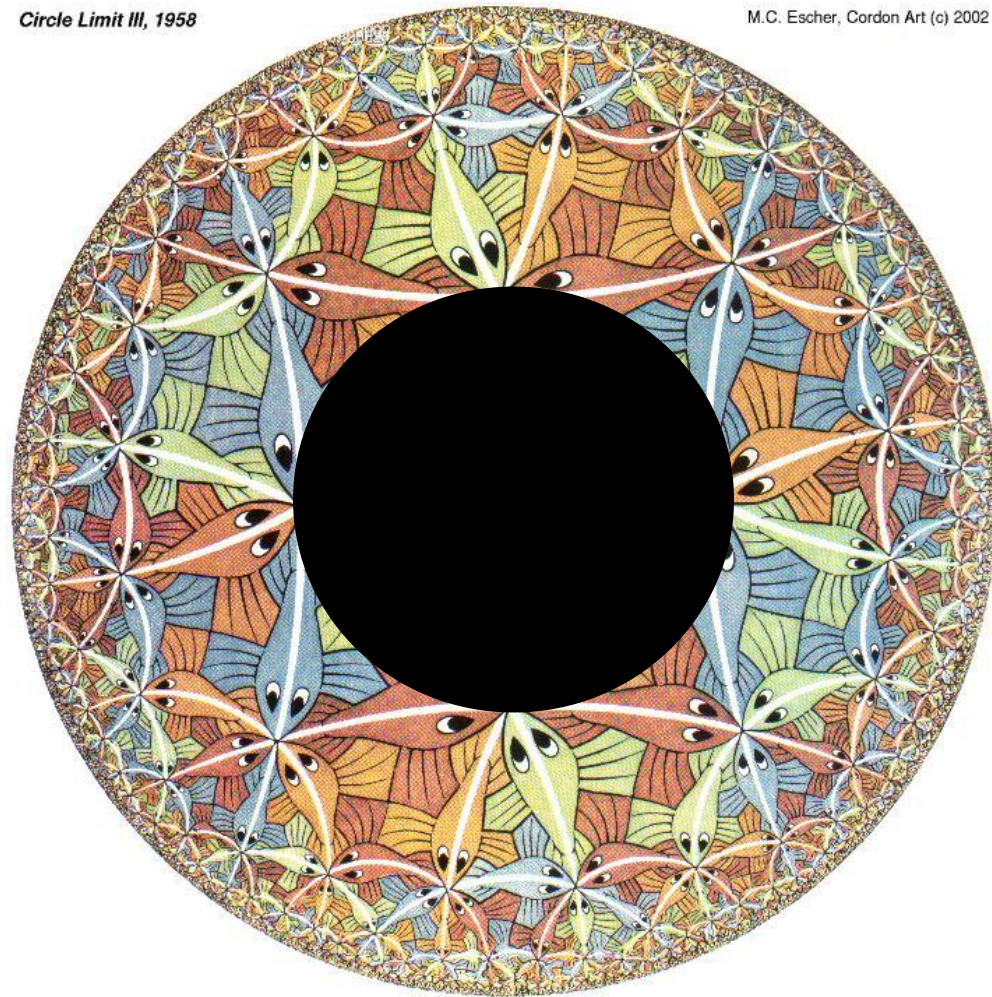
Circle Limit III, 1958

M.C. Escher, Cordon Art (c) 2002



Circle Limit III, 1958

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Entanglement

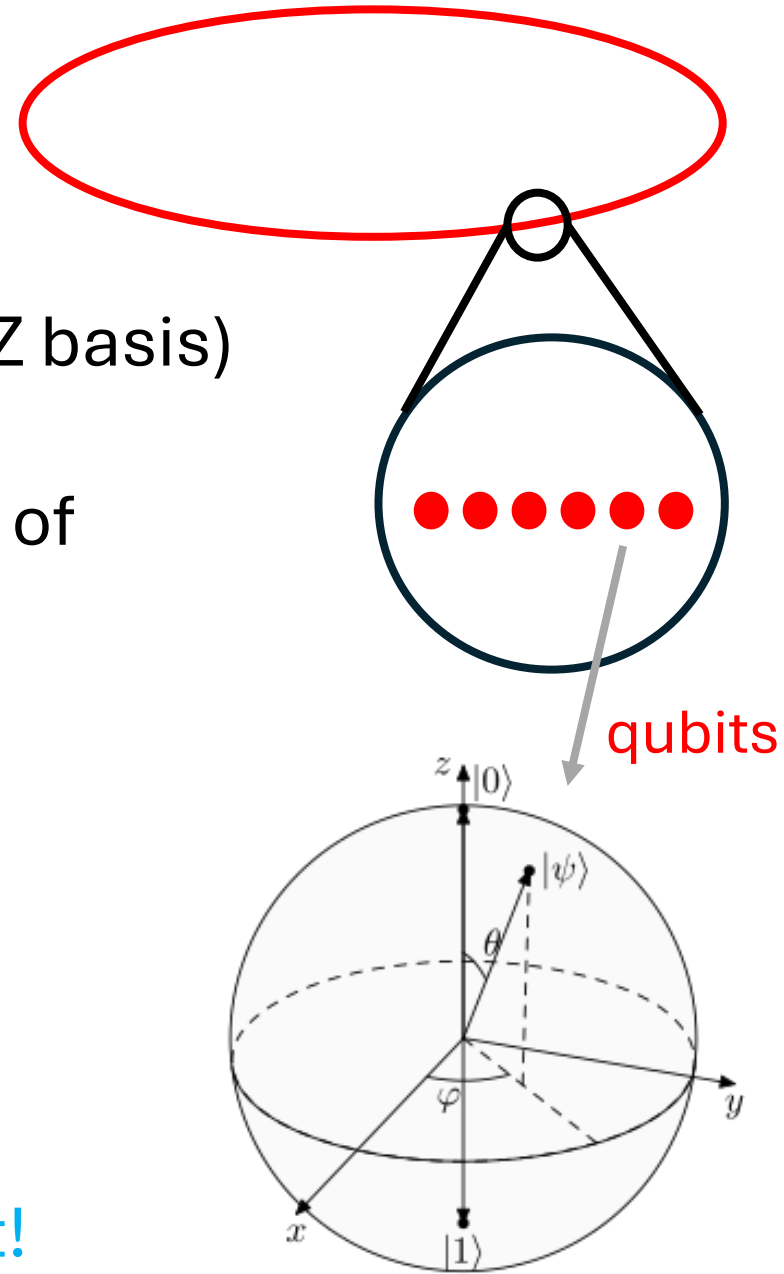
Four states: 00 and 01 and 10 and 11 (Z basis)

Superposition: any linear combination of states is also allowed

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \neq |\phi_1\rangle |\phi_2\rangle$$

EPR or Bell pair

Definition of entanglement!



Ryu-Takayanagi entanglement formula

S = Von Neumann entropy

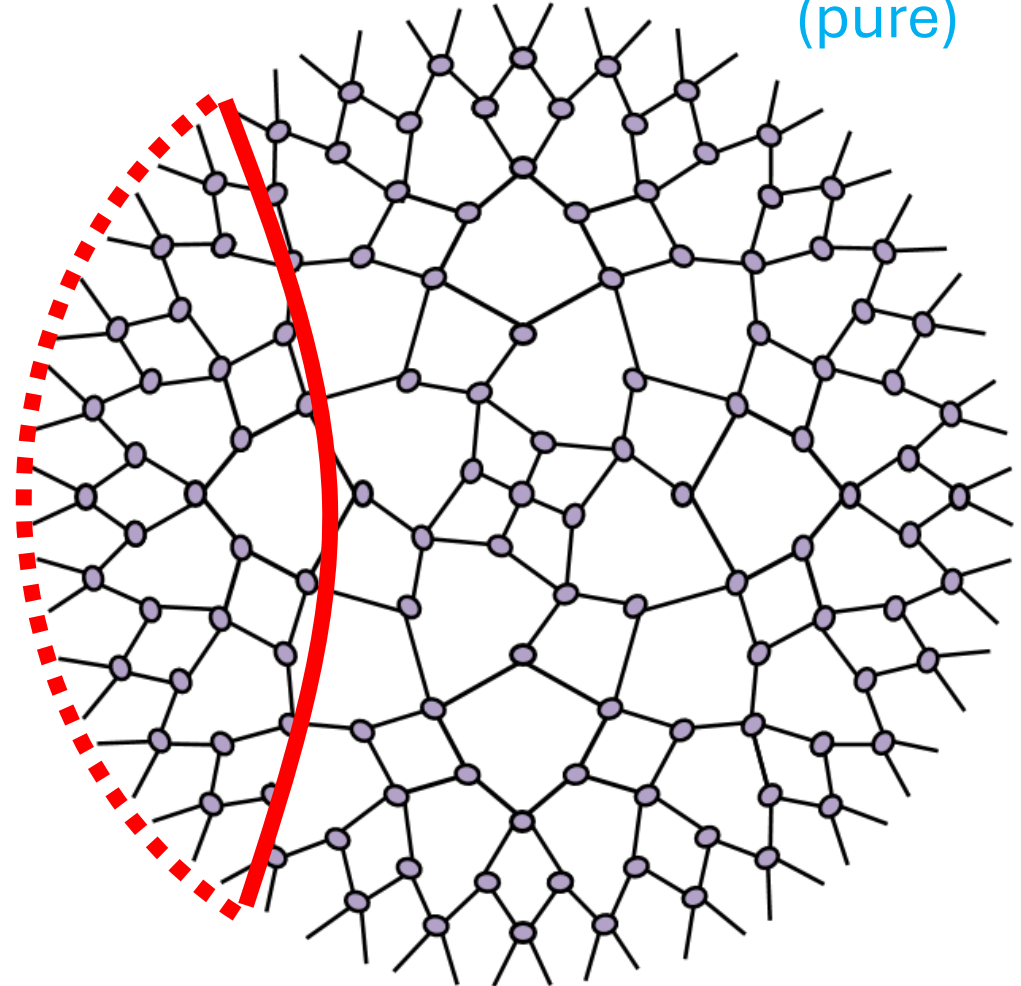
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A

$$S(A) = \frac{\text{Minimal Area}}{4\hbar G_N}$$

[Ryu-Takayanagi]

ground state
(pure)



[S, Pastawski et al., Hayden et al., ...]

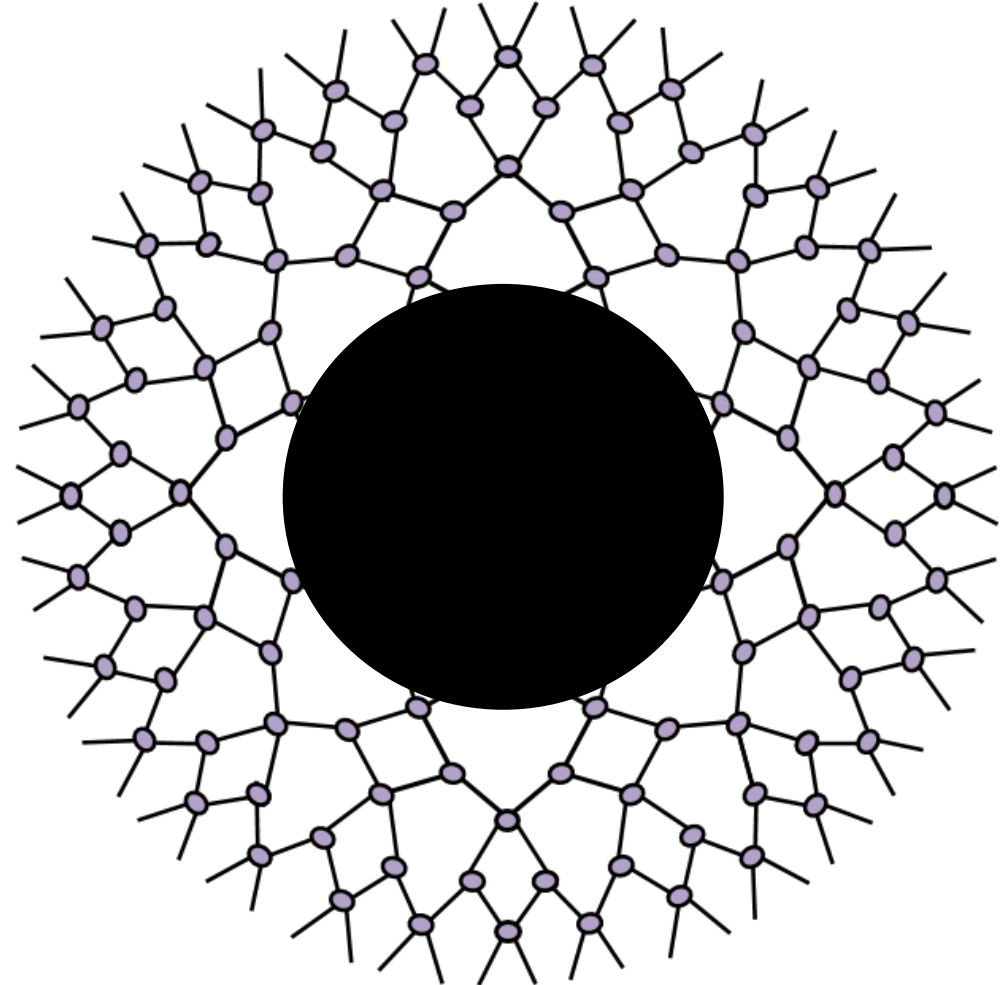
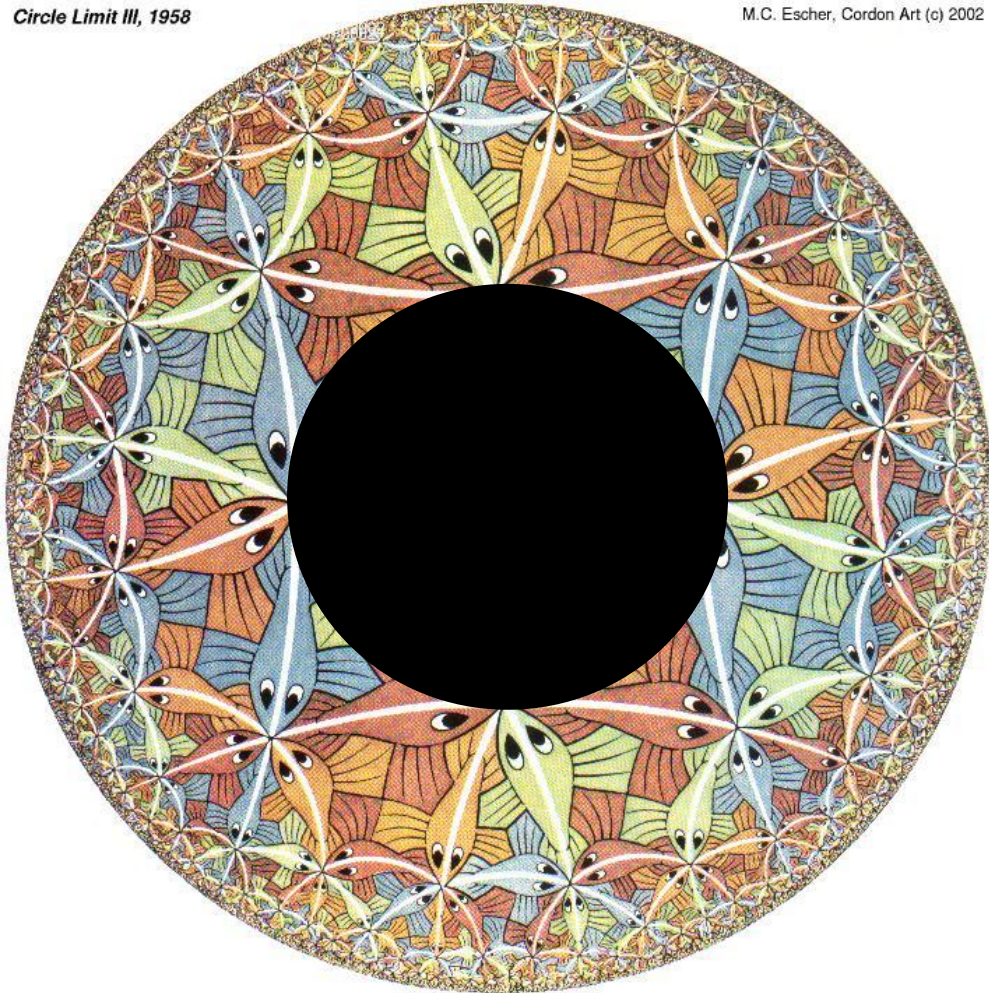
Building a black hole

$$\rho = \frac{e^{-\beta H}}{Z(\beta)}$$

thermal Gibbs
state (mixed)

Circle Limit III, 1958

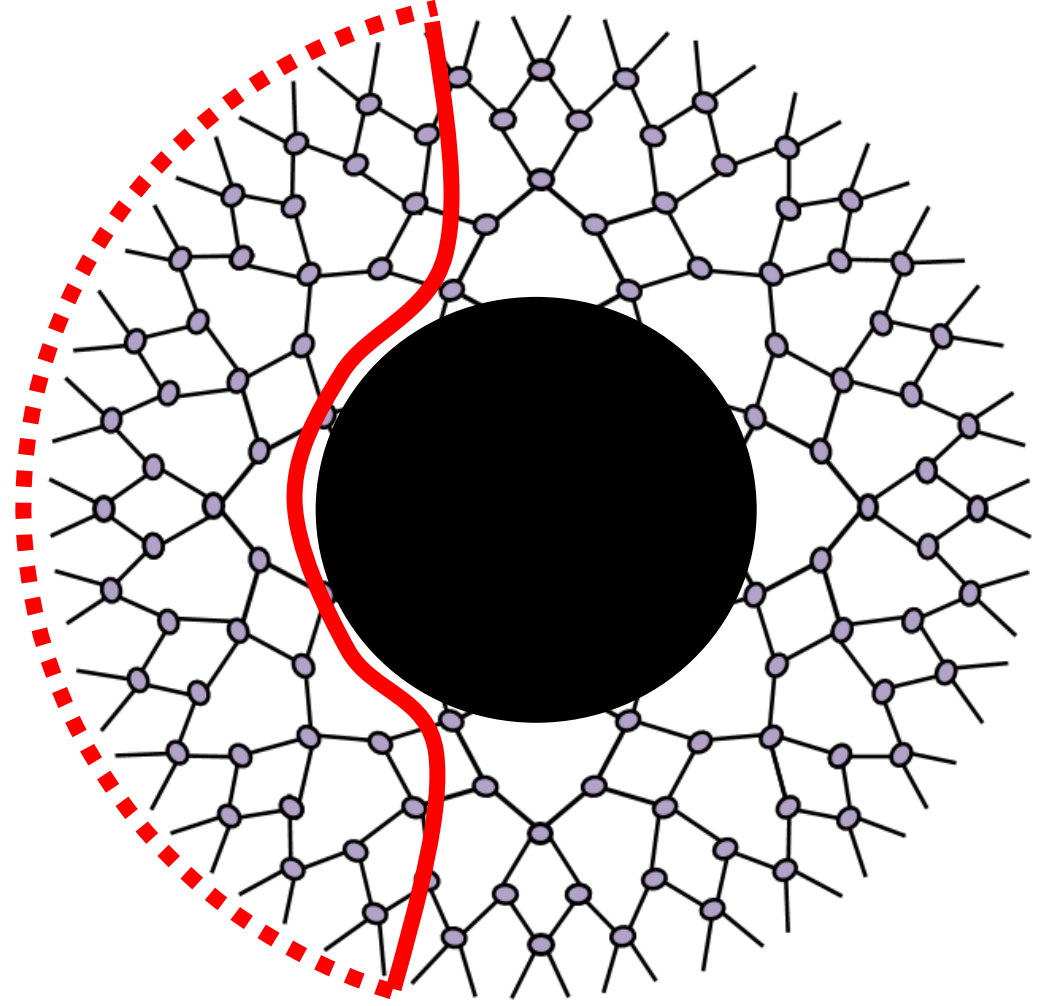
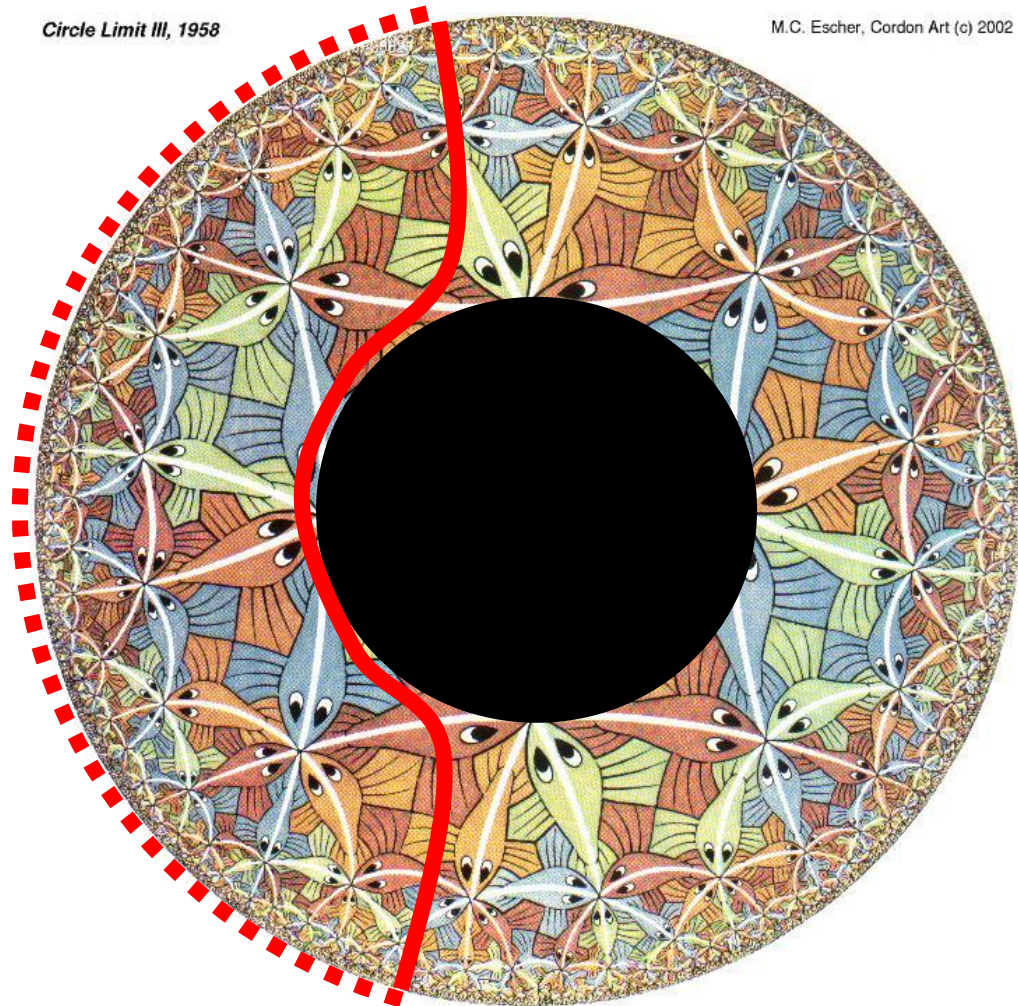
M.C. Escher, Gordon Art (c) 2002



Black hole entropy

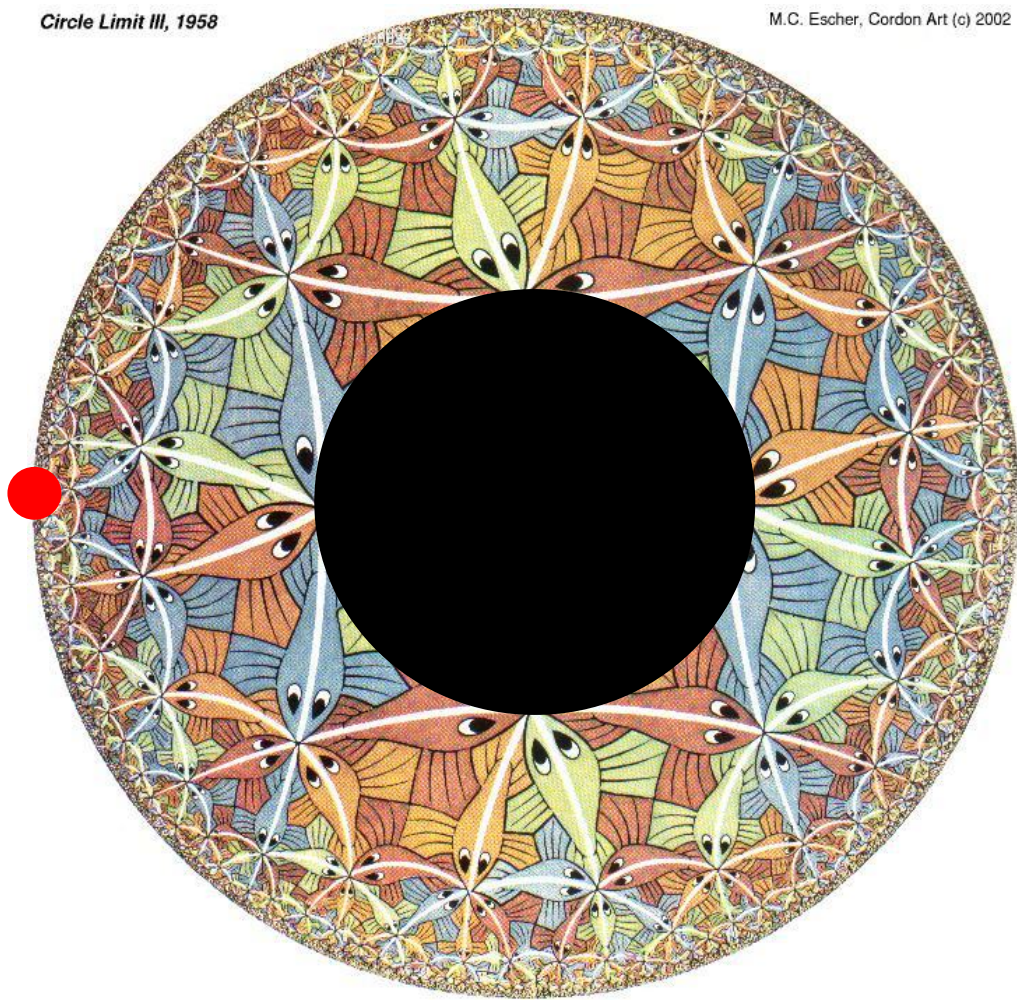
Circle Limit III, 1958

M.C. Escher, Gorden Art (c) 2002



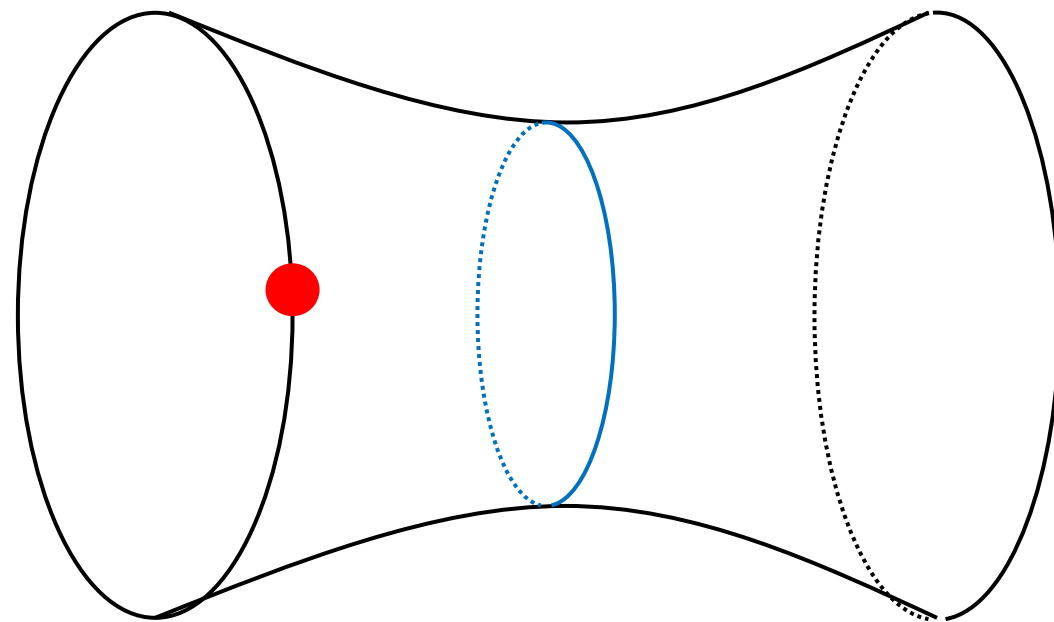
Circle Limit III, 1958

M.C. Escher, Cordon Art (c) 2002



~ mixed state

Two-sided wormhole

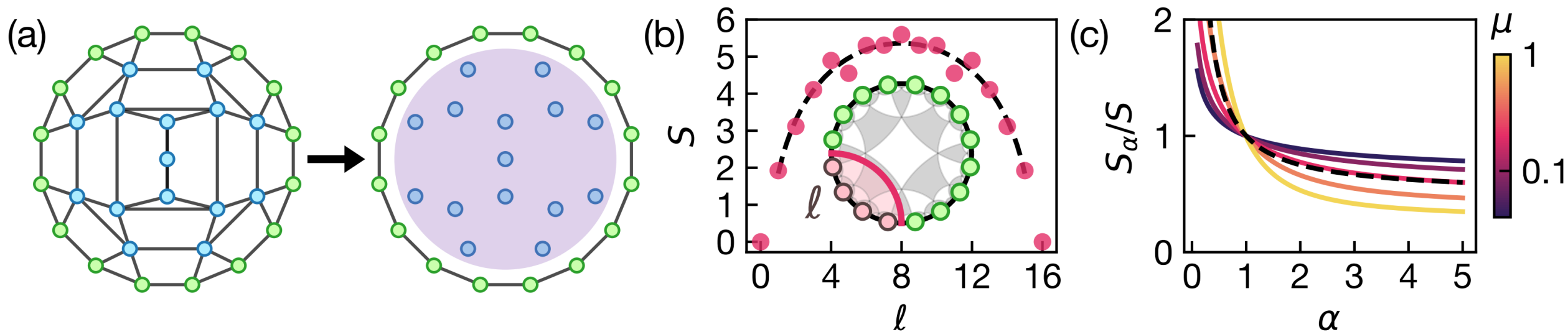


~ purification

Entanglement from geometry

Our proposal and key results

- Inspired by experimentally motivated constraints, we exhibit a simple setting with Ryu-Takayanagi-like (RT) entanglement
- Start with a graph which discretizes a chosen geometry, quench for $O(1)$ time, and measure the bulk $\rightarrow$ a pure boundary state



Key ingredients

- Degrees of freedom are sites that each contain a single canonical pair (x,p) {for example, each site may be a collection of atoms with a collective angular momentum, $[S_x, S_y] = i\hbar^2 s + i\hbar(S_z - s\hbar)$ }
- Sites have arbitrary interaction connectivity {this could be achieved by coupling atomic ensembles with light}
- Any Gaussian operations are allowed: phase space shifts, evolution with quadratic Hamiltonians, and measurements of x,p
- These ingredients are motivated with atomic ensembles in a cavity in mind, but there are other experimental platforms, e.g. photons

Setup

- The talk assumes the following protocol:
 1. Choose a graph $G=(V,E)$ with bulk and boundary sites comprising V
 2. The initial wavefunction is a product of Gaussians for each site in V
 3. The system is quenched: evolve with $H_q = \frac{1}{2} \sum_{ab} x_a J_{ab} x_b$ for unit time where J is set by the (weighted) adjacency matrix of G
 4. Measure the bulk sites in the momentum basis, and (optionally) apply local phase space displacements conditioned on the outcomes
- We analyze the entanglement entropy, Renyi entropy, and correlations of the resulting pure states

$$\xi = (x_1, \dots, x_N, p_1, \dots, p_N) \quad V_{ij} = \frac{1}{2} \langle \{ \xi_i - \langle \xi_i \rangle, \xi_j - \langle \xi_j \rangle \} \rangle$$

[literature: continuous variable graph states, methods to compute entropies]

$$e^{-\mu x_A^2/2 - \mu x_B^2/2 - iJ x_A x_B}$$



Highly squeezed limit

- Two modes (x_A, x_B) $\psi(x_A, x_B) \sim e^{iJ x_A x_B}$ highly squeezed, $\mu \ll 1$

$$\rho(x_A, x'_A) \sim \int dx_B e^{iJ x_A x_B - iJ x'_A x_B} \sim \delta(x_A - x'_A)$$

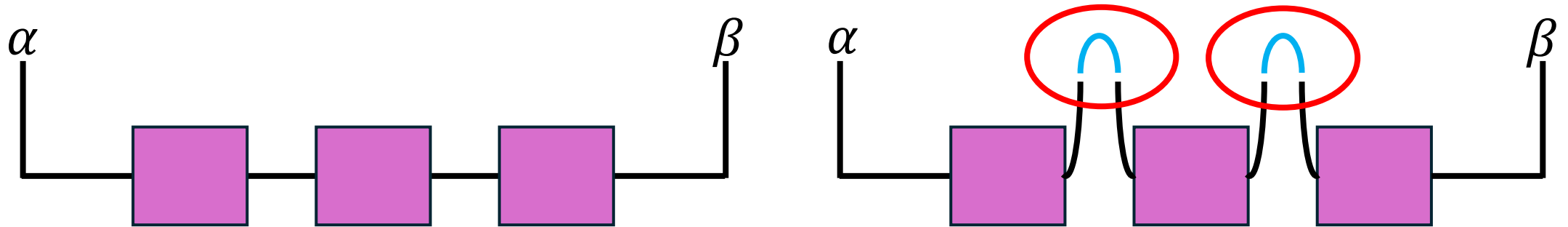
- k A-modes, k B-modes $\psi(x_A, x_B) \sim e^{\frac{i x_A J_{AA} x_A}{2} + \frac{i x_B J_{BB} x_B}{2} + i x_A J_{AB} x_B}$

$$\rho(x_A, x'_A) \sim \delta(x_A - x'_A) \text{ provided } J_{AB} \text{ is full rank}$$

[“perfect”-like tensors, see also [Kwon-Brady-Albert](#)]

Intuition

- Key inspirations: **tensor networks** and **measurements**
 - Tensor networks give a useful toy model for holography (e.g. recent MERA-based toy models of closed universe cosmology [[Antonini-Sasieta-S](#)])
 - Tensor networks also provide a powerful computational tool (e.g. TMERA [[Sewell-White-S](#)], forthcoming work on chiral clock model [[Guo-S](#)])
 - Measurements, when combined with feedforward, can yield efficient preparations of states (e.g. [literature on topological state prep](#))

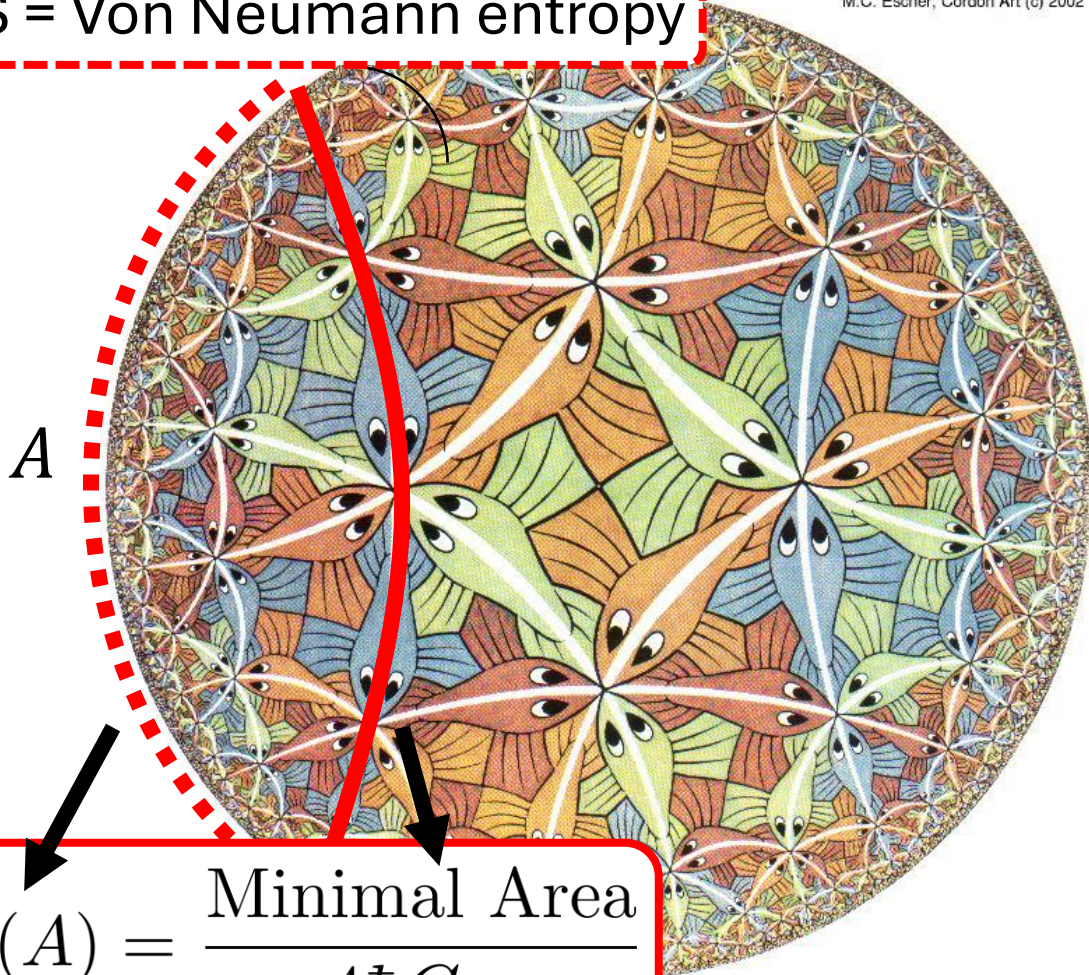


Implement internal contractions
by projecting onto Bell pairs

Example 1: hyperbolic disk

S = Von Neumann entropy

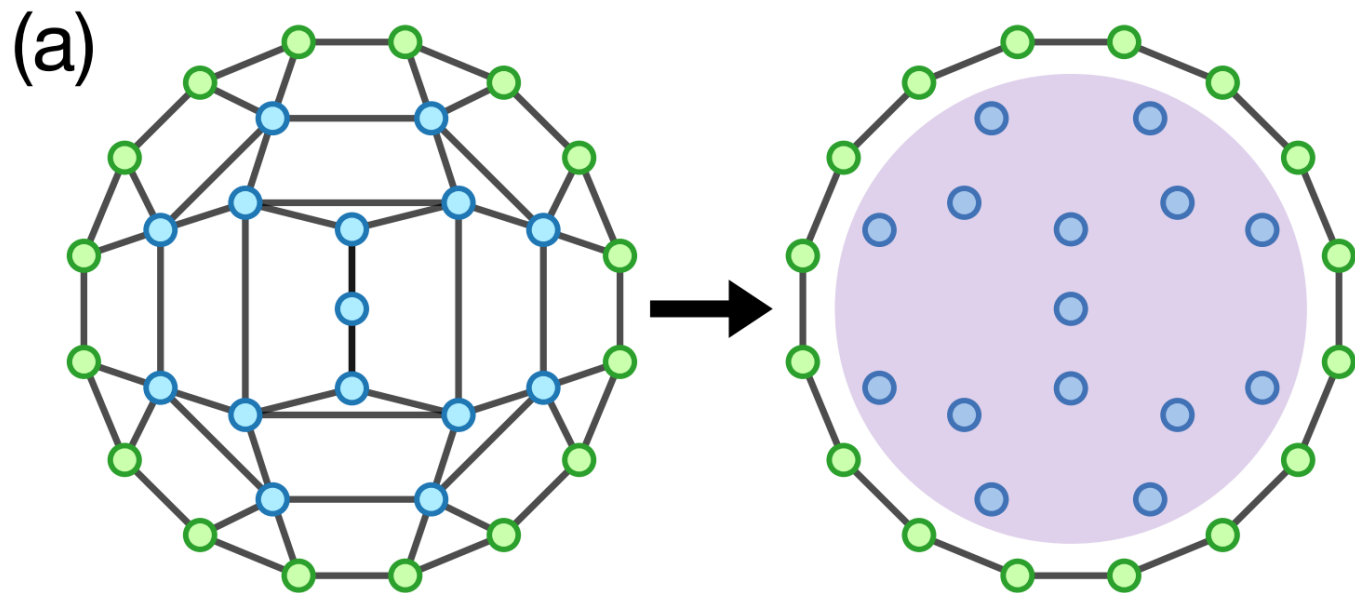
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- **AdS/CFT**: the ground state of the boundary (1+1)d CFT is dual to empty (2+1)d AdS space, spatial geometry is a hyperbolic disk
- The RT surface (curve in 2+1) is a portion of a circle (set $\hbar = 1$)

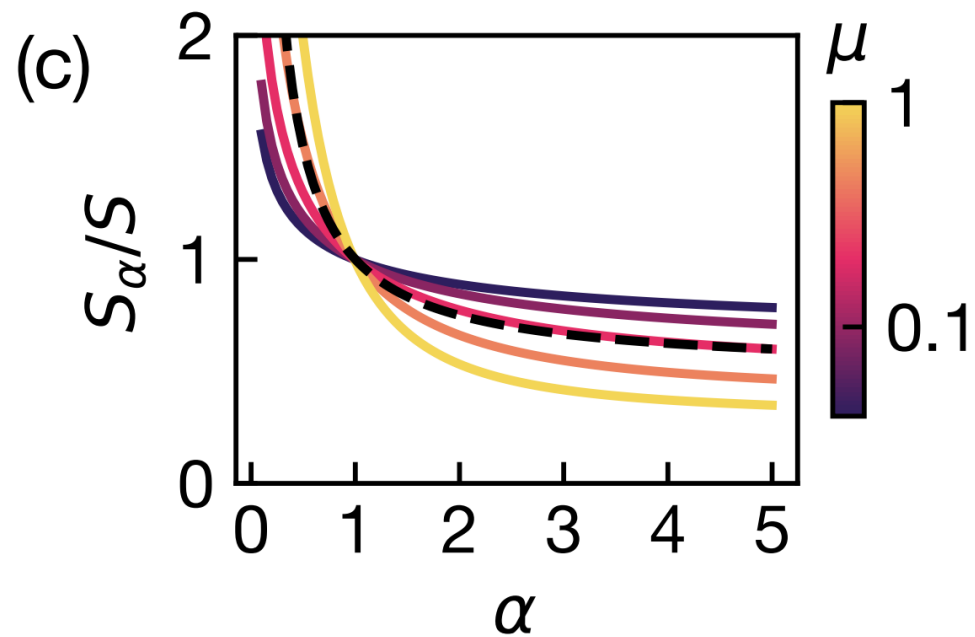
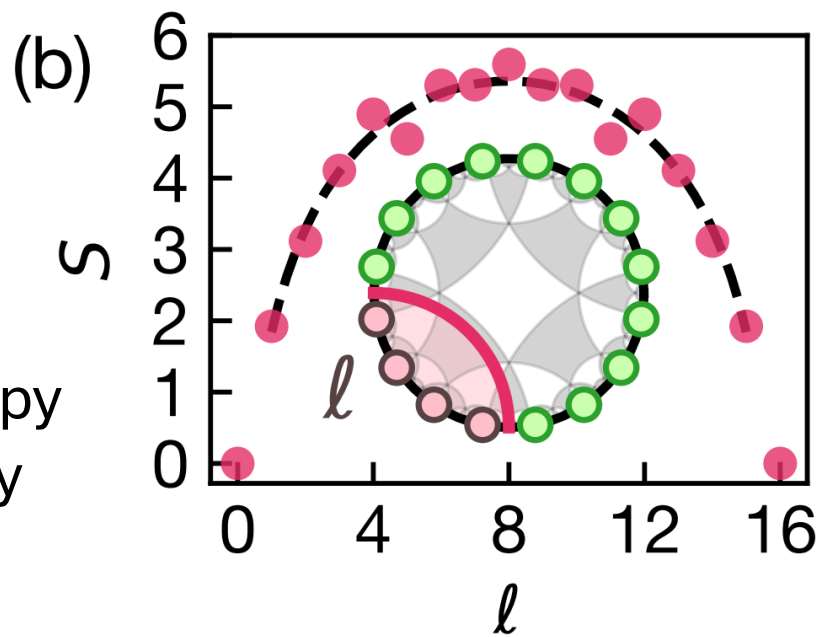
$$S = \frac{d_{\text{bulk}}}{4G} = \frac{\ell_{\text{AdS}}}{2G} \ln \left(\frac{2r_c}{\ell_{\text{AdS}}} \sin \frac{\Delta\theta}{2} \right)$$

$$S(A) = \frac{\text{Minimal Area}}{4\hbar G_N}$$



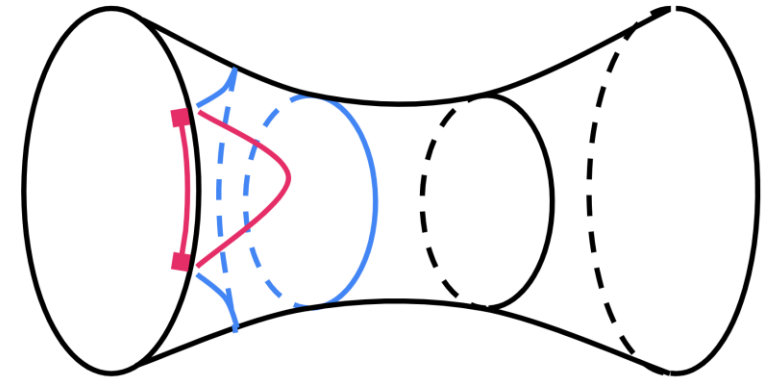
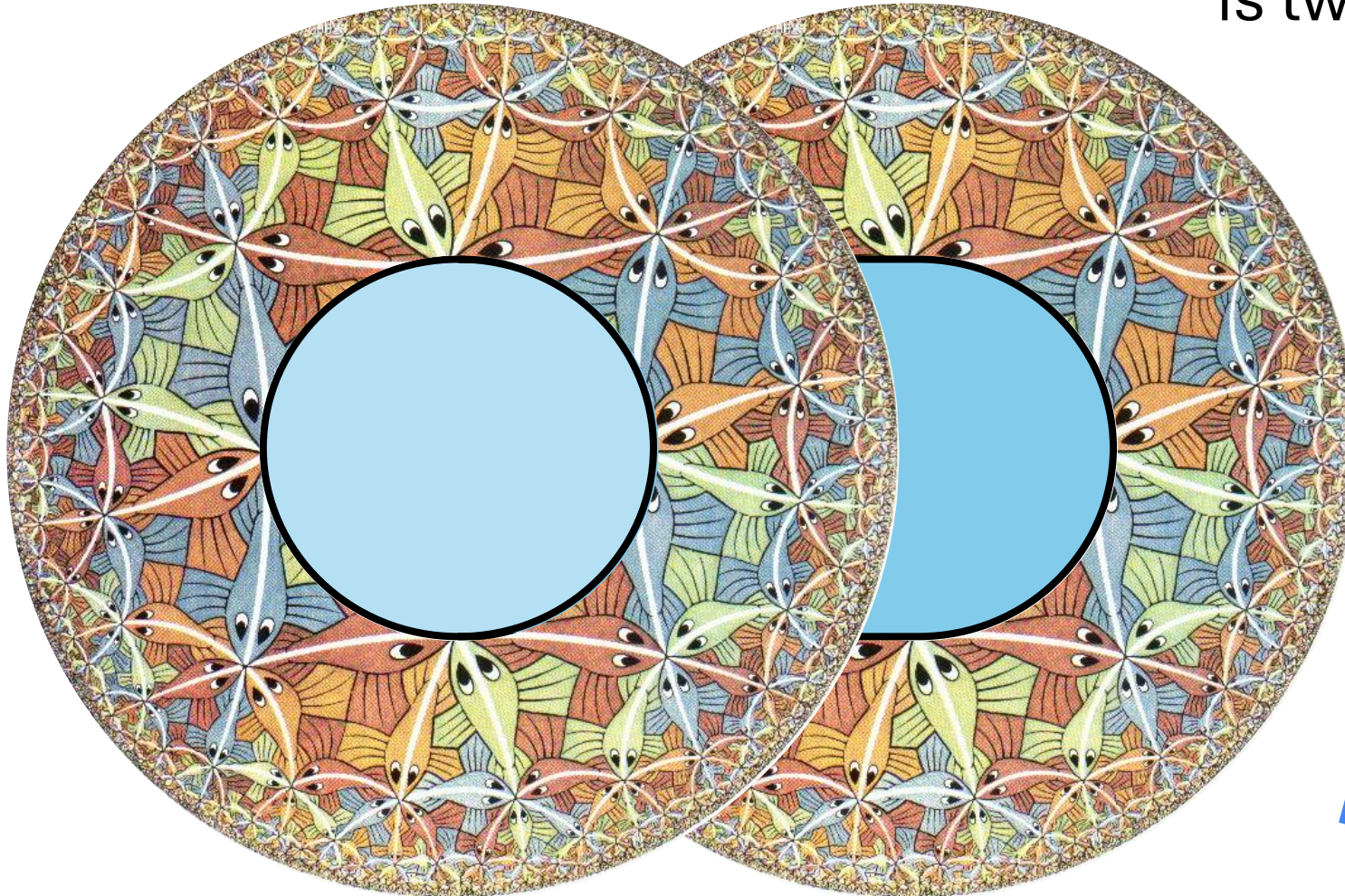
A simple discretization of the hyperbolic disk, followed by measurement of the bulk sites

(b) Entanglement entropy $\mu = .2$, (c) Renyi entropy as a function of μ



Example 2: wormhole

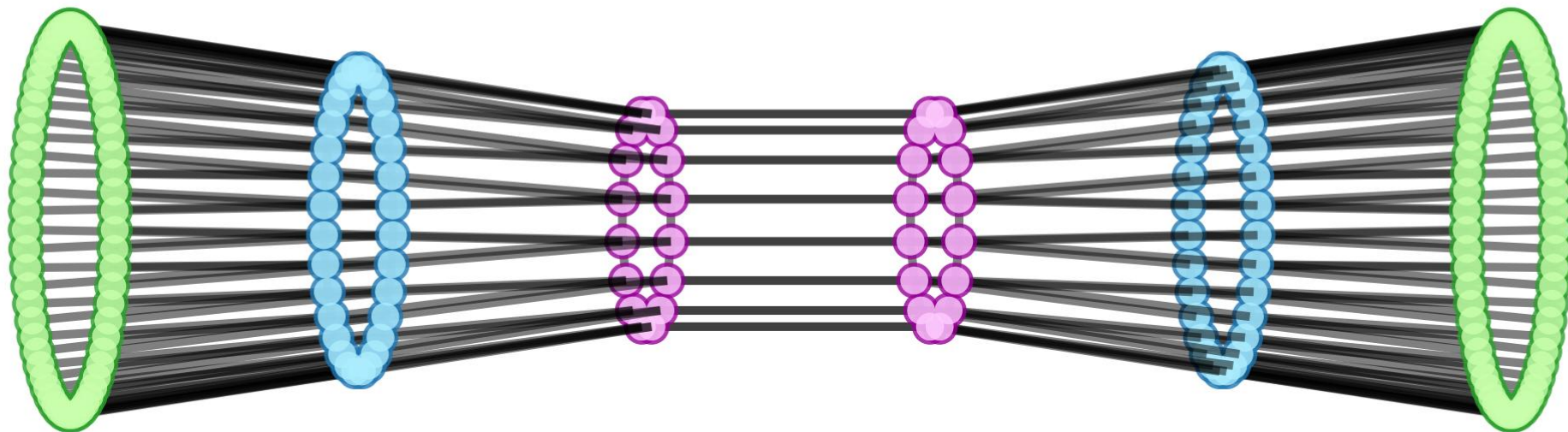
- **AdS/CFT:** a thermofield double state of the boundary (1+1)d CFT is dual to a wormhole connecting two black holes, spatial geometry is two annuli connected by a tube



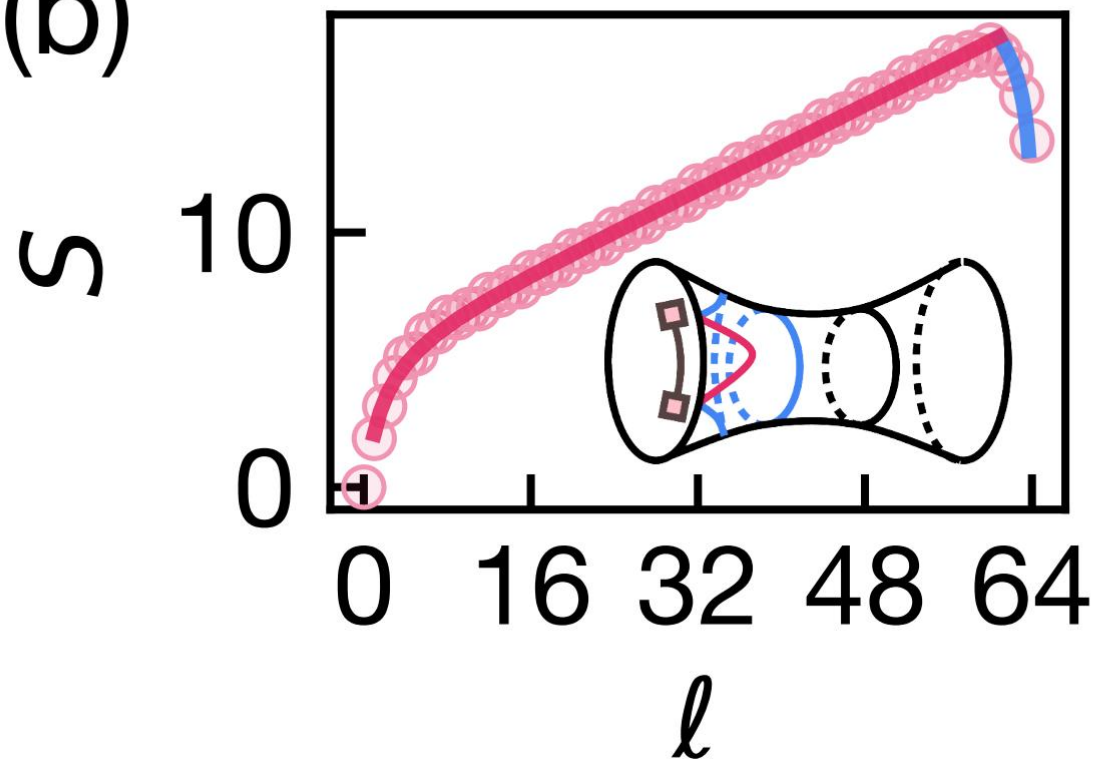
$$/ S^{(1)} = \frac{d_{\text{bulk}}}{4G} = \frac{c}{3} \ln \left(\frac{2r_c}{r_+} \sinh \frac{r_+ \Delta\theta}{2\ell_{\text{AdS}}} \right)$$

$$/ S^{(2)} = S_{th} + \frac{c}{3} \ln \left(\frac{2r_c}{r_+} \sinh \frac{r_+ (2\pi - \Delta\theta)}{2\ell_{\text{AdS}}} \right)$$

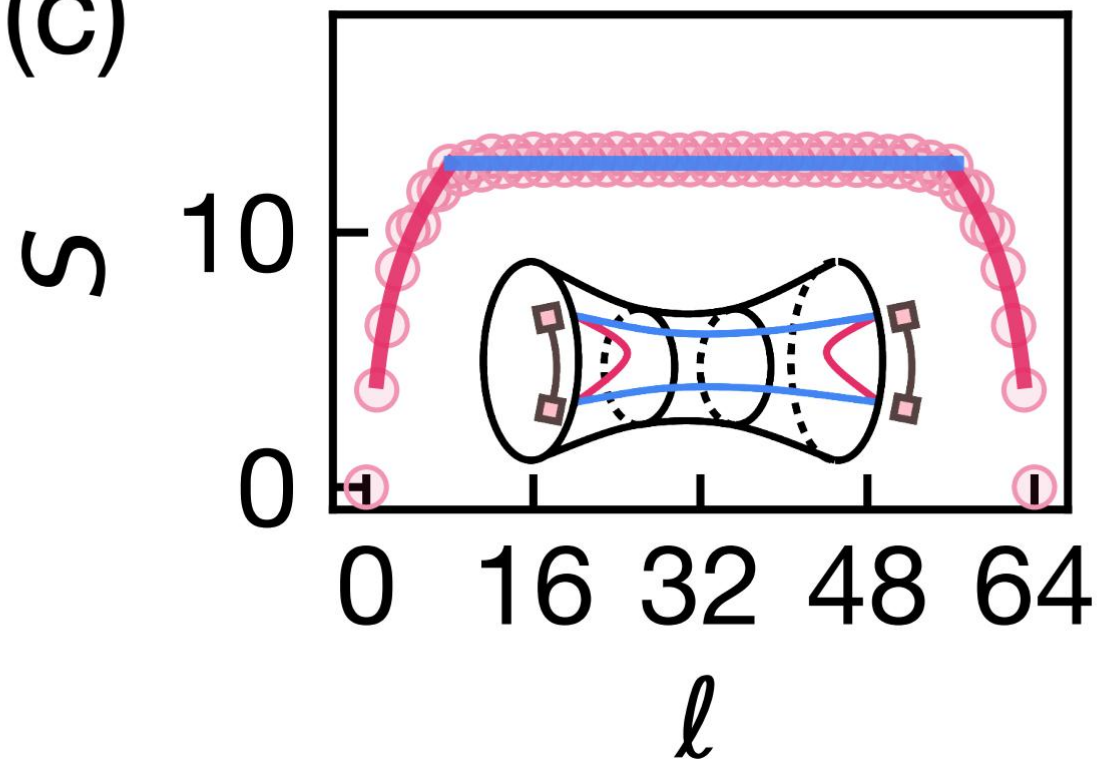
(a)



(b)

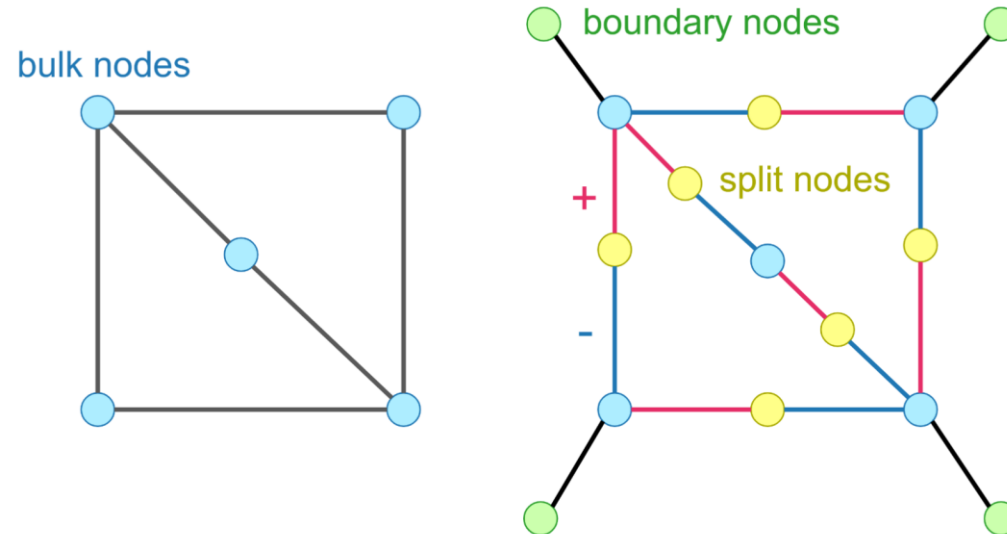


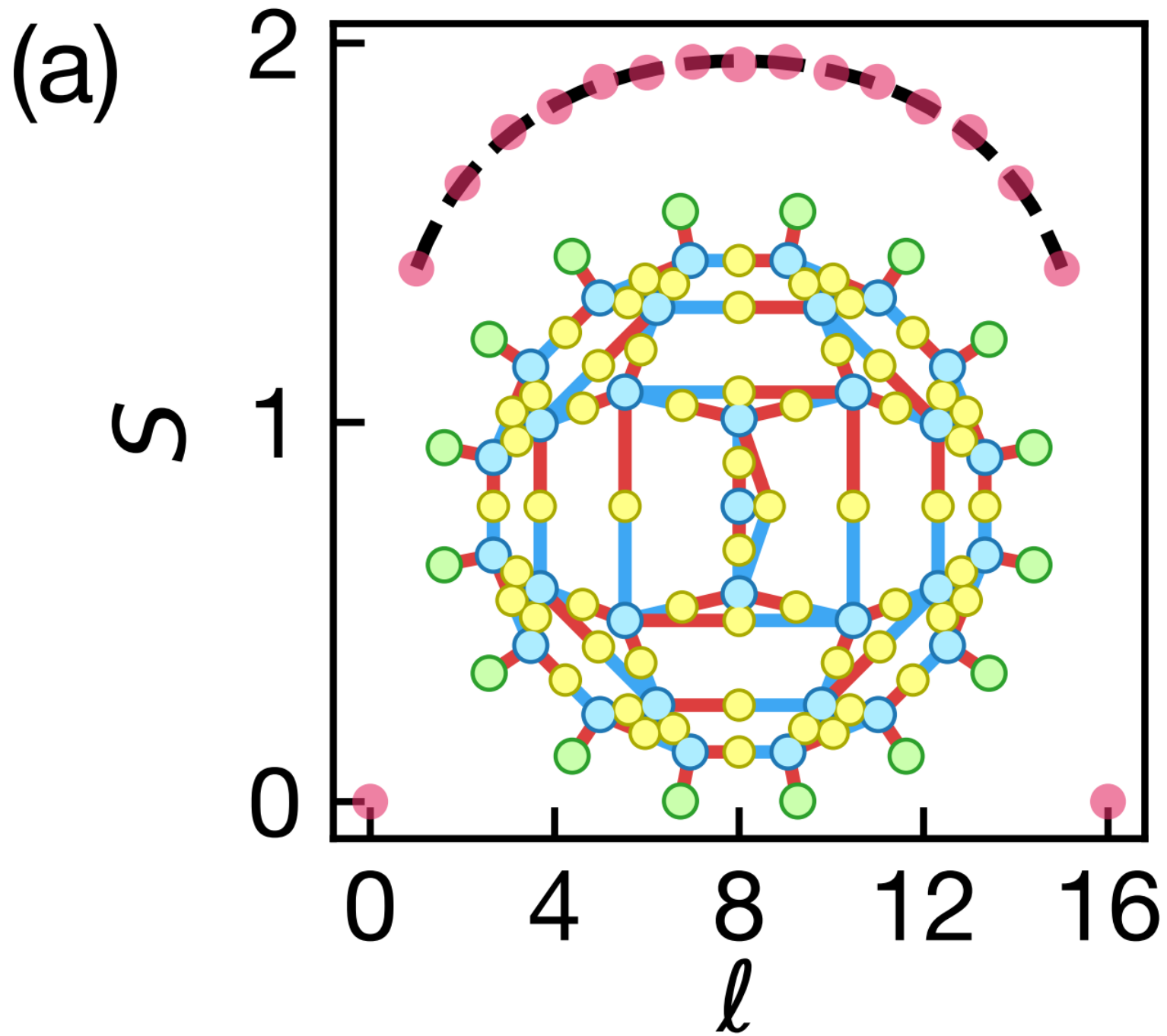
(c)

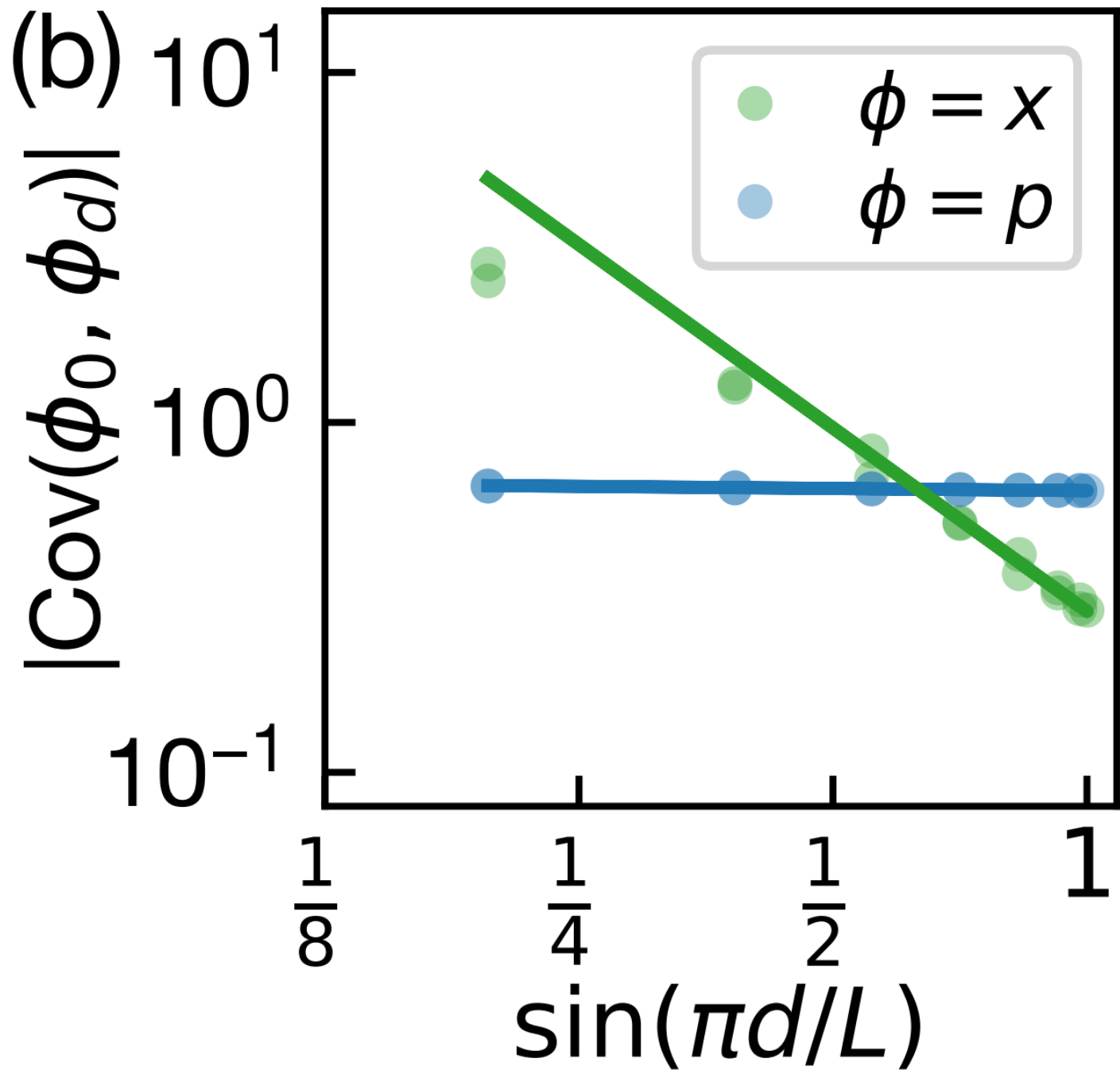


Example 3: decorated hyperbolic disk

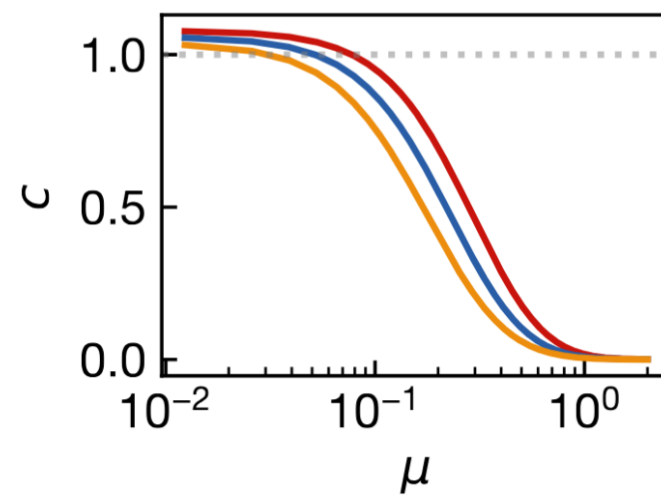
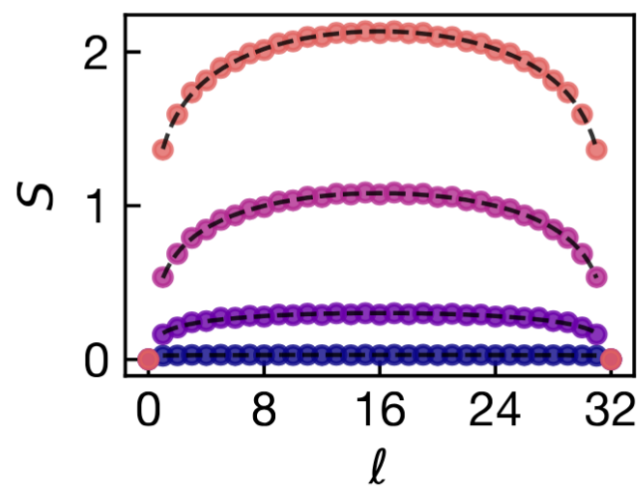
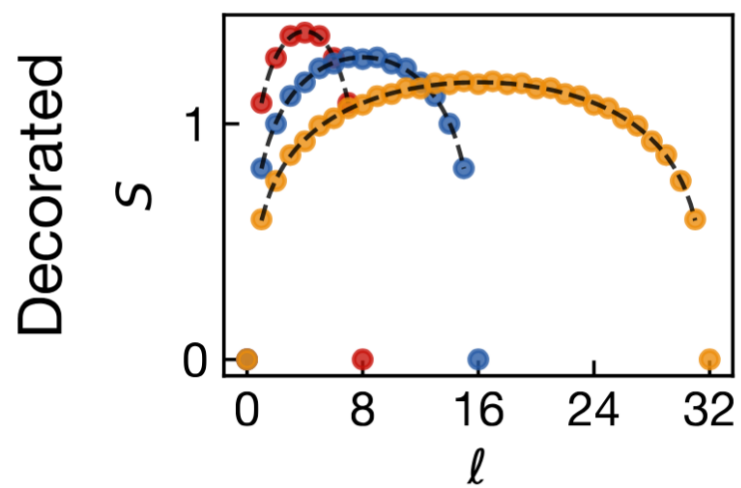
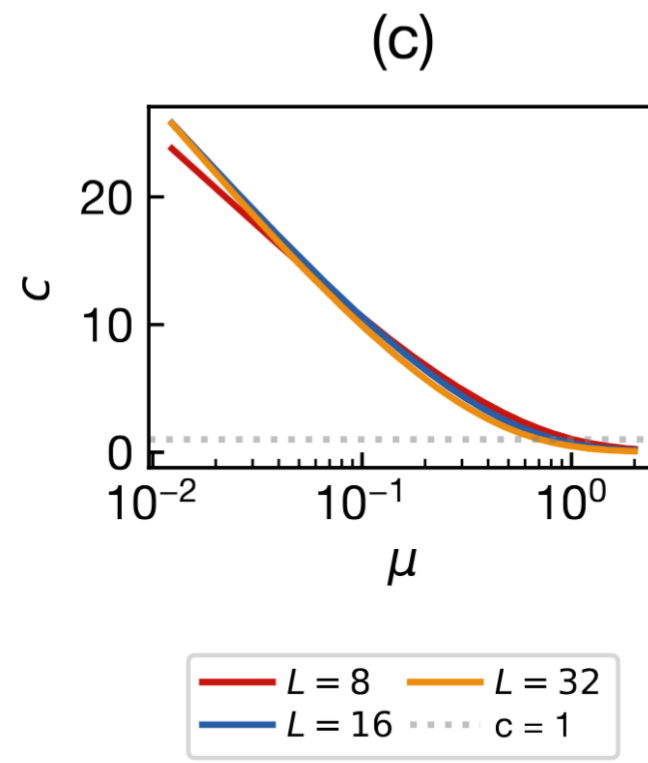
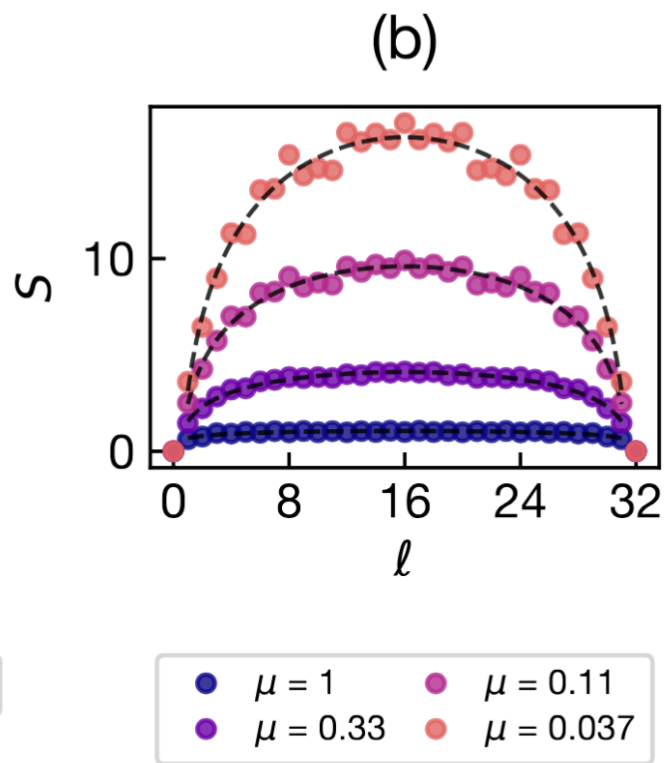
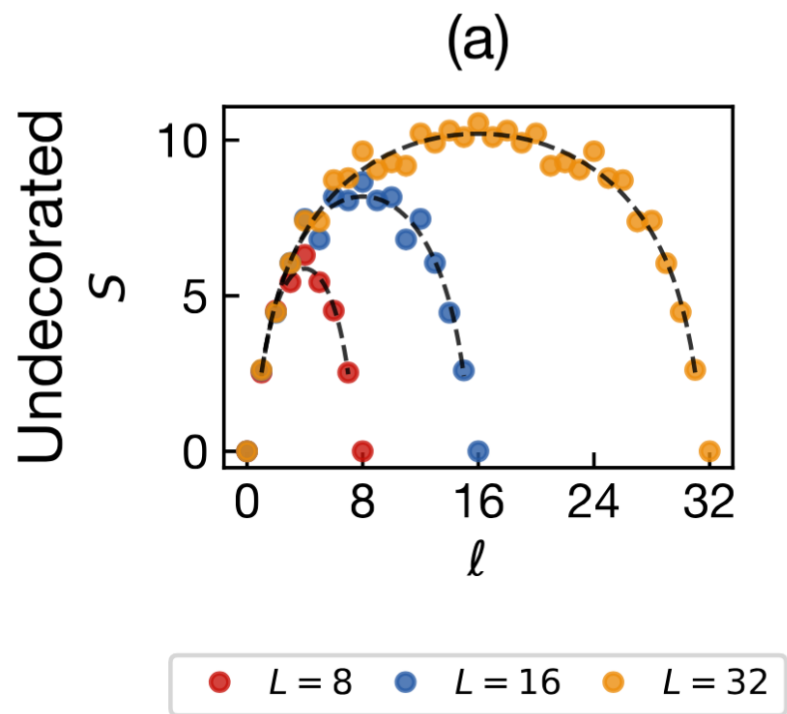
- We've achieved RT-like entanglement entropy and matched CFT Renyi entropy, at least for single intervals in the ground state
- We also wanted to obtain power-law correlations, which turns out to be possible using a decoration procedure







Scaling dimensions ($\approx 0,1$), central charge (≈ 1), and Wick's theorem are close to our expectations for a free boson CFT ground state. [Did we build this CFT state at long wavelengths?](#)

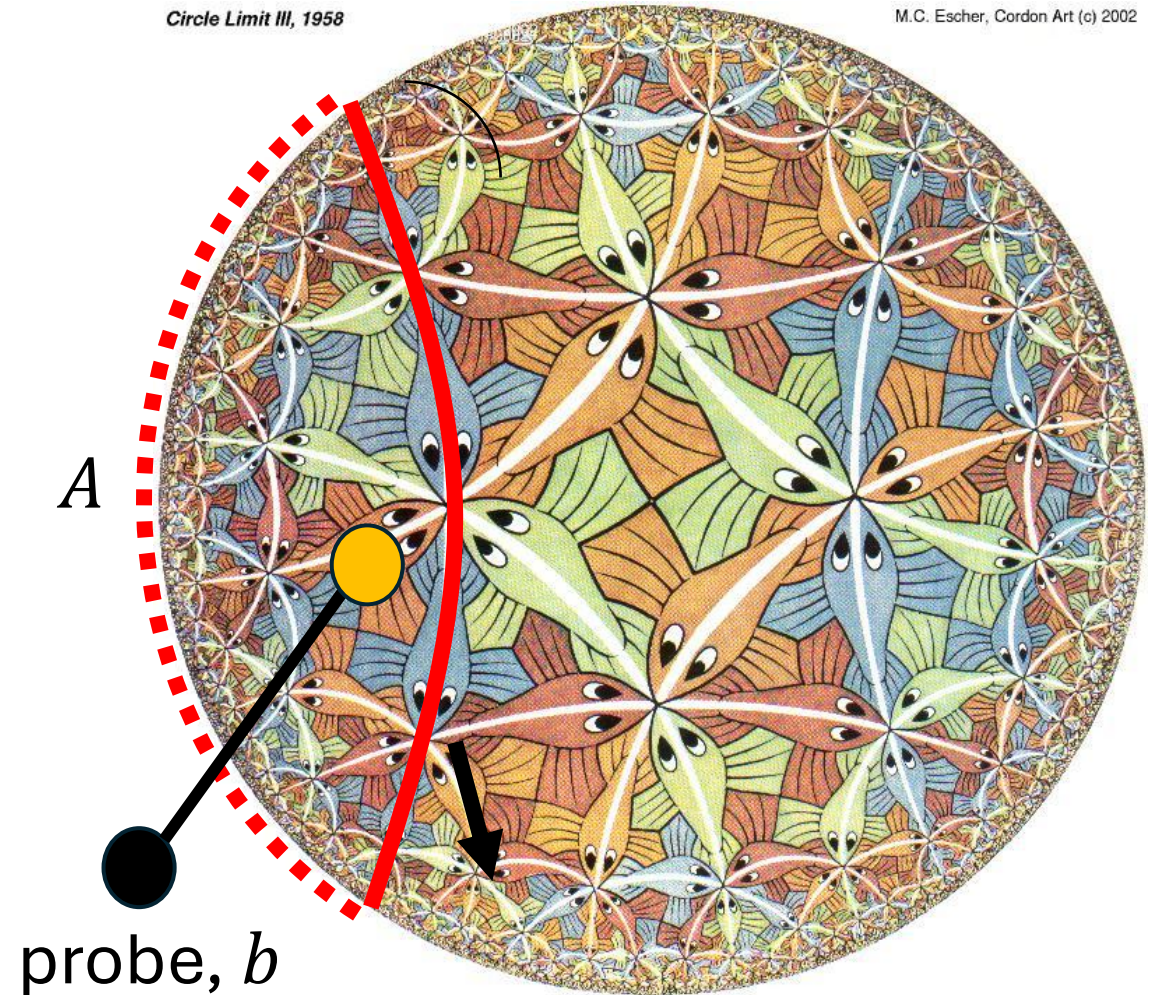


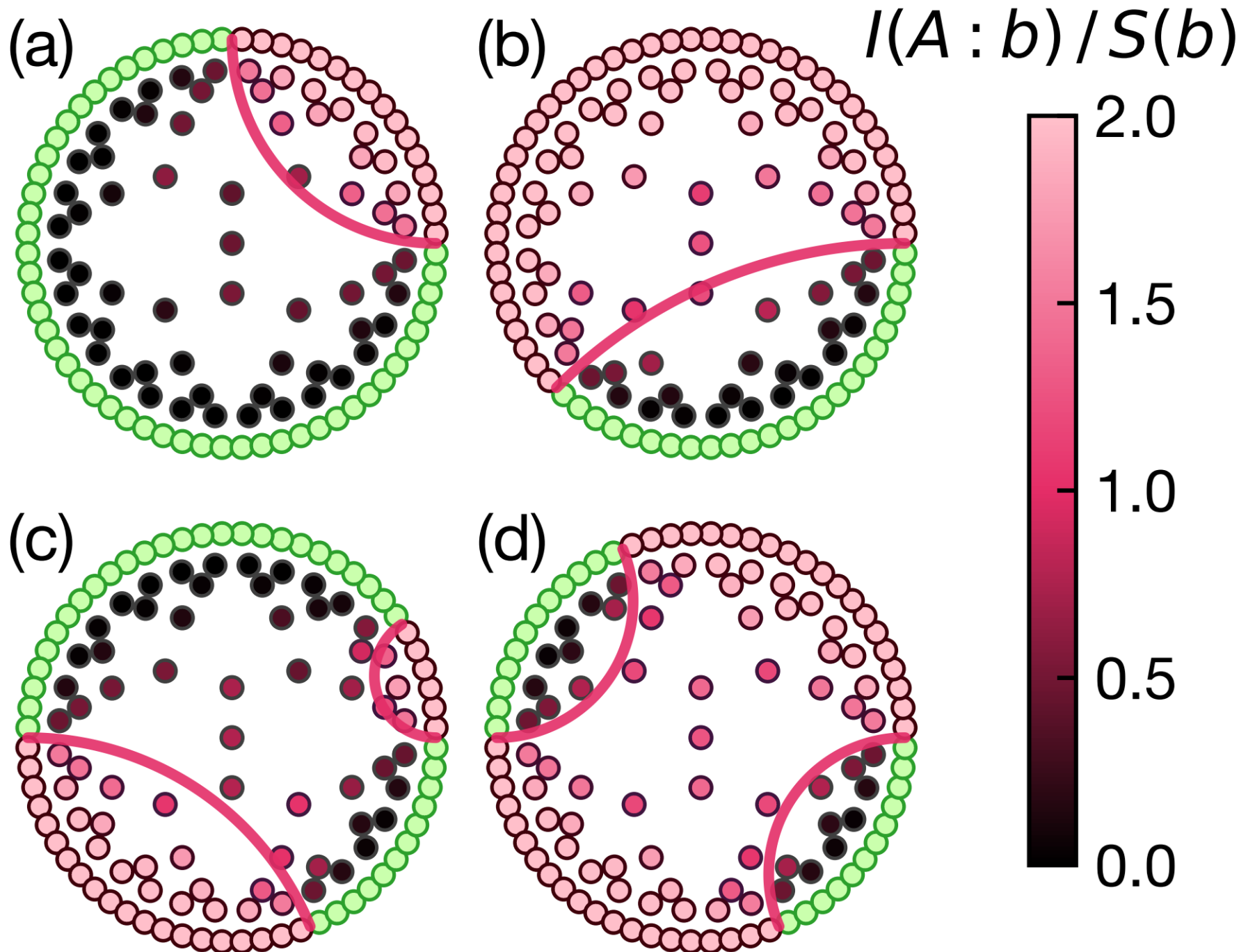
Bulk reconstruction

- We'd like to directly detect the RT surface from data
- Add one extra site, the probe, which is unmeasured and directly connects to a chosen bulk site ●
- AdS/CFT: $I(A: b)/S(b) = 2$ if probe connects inside the RT surface of A and $= 0$ if it connects outside

$$I(A: b) = S(A) + S(b) - S(Ab)$$

mutual information





Outlook

There are a lot of potential directions, including:

- We want to think about adding non-Gaussianity and dealing with post-selection [e.g. backcreation [Cao et al. 2403.07056?](#)]
- Experimental implementation is also crucial, and I think the quantum extremal surface formalism will help us model the effects of imperfections, e.g. incomplete bulk measurements
- The wormholes can be viewed as TFD states of some quadratic Hamiltonian, so we can study teleportation dynamics ([WIP Allison Powell, Brian Khor](#))

Perturbing the hyperbolic disk $g_{ij}^{(0)} \mapsto g_{ij} = g_{ij}^{(0)} + h_{ij}$

- Tensor Radon transform: $R_2[h_{ij}](\gamma_A) \equiv \int_{\gamma_A} h_{ij} \dot{\gamma}_A^i(s) \dot{\gamma}_A^j(s) ds$

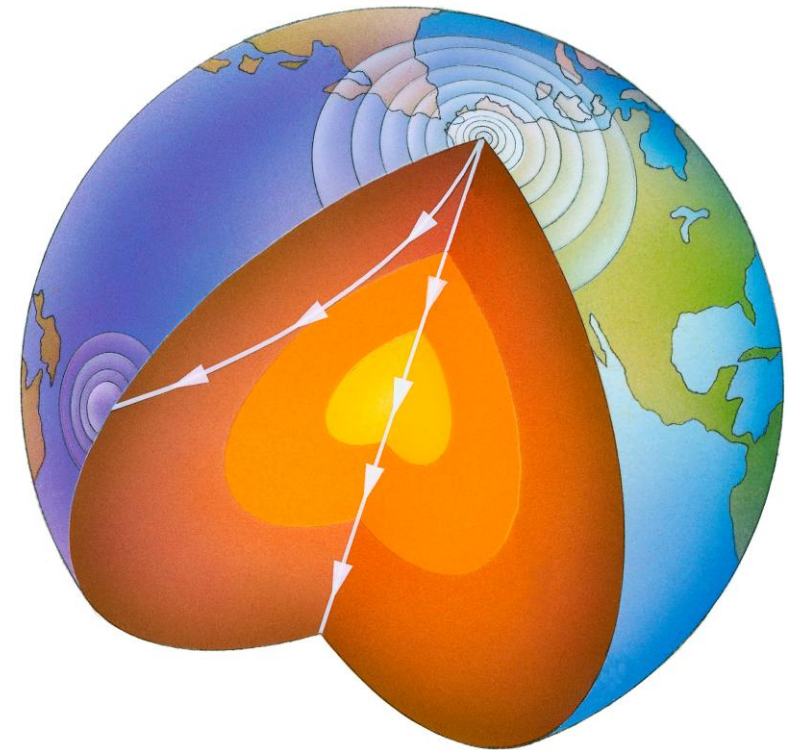
- Idea: invert transform to determine metric perturbation from entanglement perturbation

- Must be careful about gauge-invariance

$$R_2[\nabla_i \xi_j + \nabla_j \xi_i] = 0$$

- Related problems appear in geophysics, where one wants to learn about the properties of the Earth's interior from seismic wave timing data

[Tang-Cao-Qi-S 2007.00004]



[from DK Find Out website]

Algorithm

- Discretize the bulk geometry into tiles with piecewise constant metric
- Radon transform takes the form of a linear map from space of metric perturbations to space of entanglement perturbations

$$2\Delta L(\gamma_A) = R_2[h_{ij}] \approx \sum_{\mathcal{T}} W^{ij}(\mathcal{T}, \gamma_A) h_{ij}(\mathcal{T})$$

- Invert this map within a gauge fixed sector of metric perturbations

$$\min_{\mathbf{h}} \|\mathbf{W}\mathbf{h} - \mathbf{b}\|, \quad \mathbf{h}_* \text{ minimizer}$$

subject to $\mathbf{C}\mathbf{h} = \mathbf{0}$. gauge fixing constraints

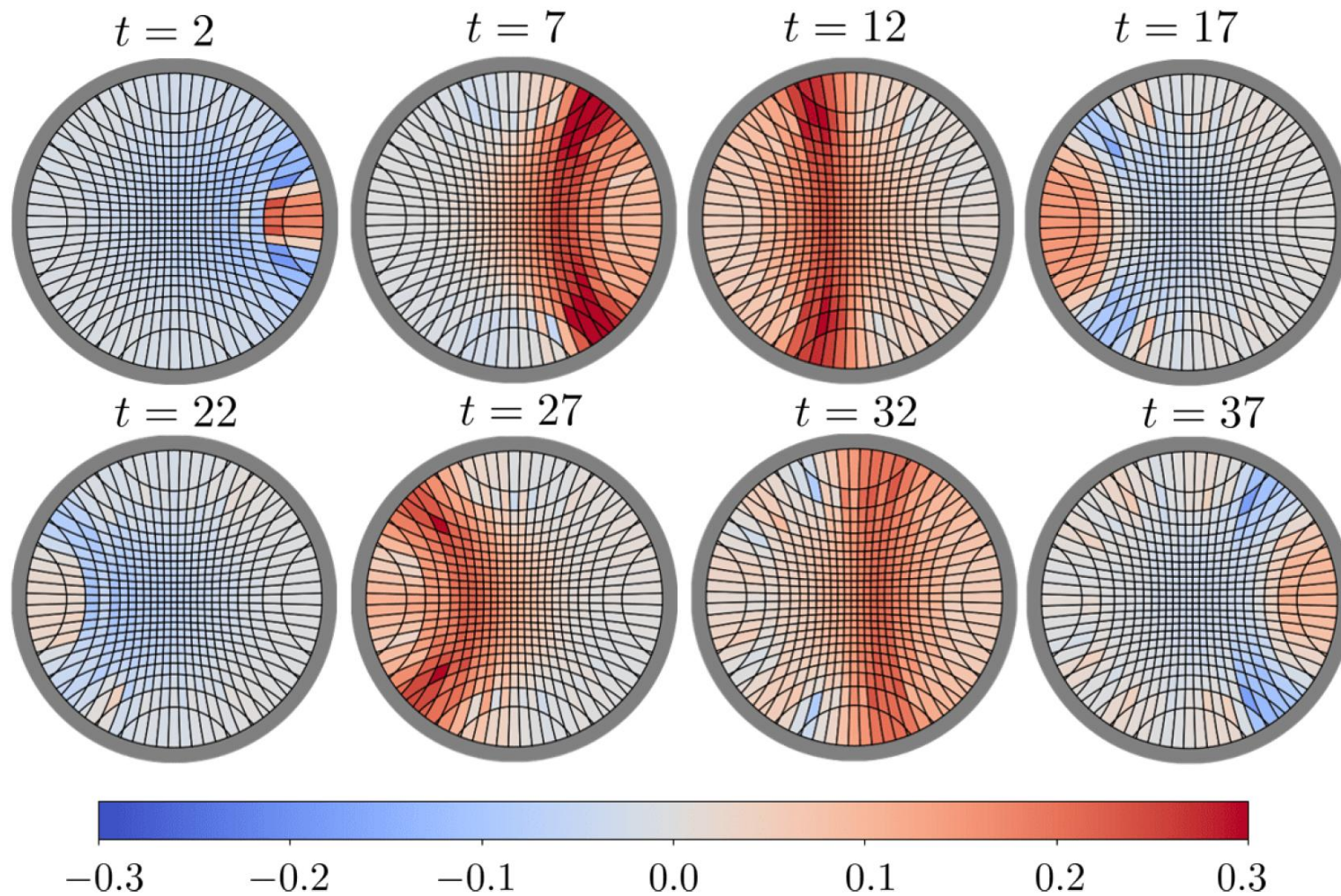


Figure 6: Quenched time dynamics of the reconstructed geometry corresponding to a single mass deformation along the x -axis. The boundary data is pre-processed by smoothing to remove small-scale details (see Figure 5 for the corresponding unsmoothed reconstruction). Only the dominant component h_{11} is shown here. The relative error is $0.03 \lesssim \mathcal{E}_{\text{bdy}} \lesssim 0.1$.

Gravitational dynamics?

- In 2014, Van Raamsdonk and I showed that linearized Einstein's equations and the equivalence principle emerged from:
 - (i) the RT formula + quantum corrections
 - (ii) CFT symmetries
- The setting was the same as the just discussed problem of metric reconstruction, namely perturbations around AdS
- We've also seen RT and CFT-like behaviors earlier in the talk
- Our protocol for building holographic entanglement by measurement seems like an excellent starting point for exploring these issues in the lab! How much non-Gaussianity do we need?

